

Treatment Assignment Design Evaluation for Spatial CRTs

Simulation Study with Heterogeneous Incidence — Results Summary

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1 Introduction

This report summarizes a simulation study evaluating **eight treatment assignment designs** for Spatial Cluster Randomized Trials (CRTs) when outcome incidence is heterogeneous and spatially structured. The central research question is:

Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment strategy minimizes estimation error for the direct intervention effect $\hat{\tau}$ across a range of spatial dependence and spillover configurations?

Application context: Sudden Unexpected Death (SUD) in North Carolina counties. The Poisson incidence mode uses an SUD base rate of 35 per 100,000 (Mirzaei et al.), consistent with county-level analyses by Gan et al. and Watson et al.

The primary estimator is Maximum Likelihood Estimation via a spatial autoregressive model (MLE), fit to the 2026-09 revised simulation. Difference in Means (DIM) appears only as a naive baseline, and its results are **pre-revision** (March 2026 run: old data-generating process, 6 designs), so DIM and MLE numbers are not like-for-like. Queen contiguity is the primary neighbor structure; rook is a sensitivity analysis. The primary MLE results use the $\tau = 1.0$ scenario (2,560 parameter scenarios); a tau sensitivity analysis across $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ (12,800 total scenarios) is presented in Section 9.

2 Mathematical Framework

2.1 Study Region and Spatial Structure

The simulation operates on a **regular lattice grid** $\mathcal{G} = \{s_1, \dots, s_N\}$ on $[1, G] \times [1, G]$, with $N = G^2$ clusters ($G = 10$, $N = 100$).

Two contiguity structures define neighbor relationships:

- **Rook contiguity:** $s_j \in \mathcal{N}(s_i)$ iff s_i and s_j share an edge (up to 4 neighbors)
- **Queen contiguity:** $s_j \in \mathcal{N}(s_i)$ iff s_i and s_j share an edge or corner (up to 8 neighbors)

The spatial weight matrix $\mathbf{W}$ has entries $w_{ij} = |\mathcal{N}(s_i)|^{-1}$ if $s_j \in \mathcal{N}(s_i)$ and 0 otherwise (row-standardized).

2.2 Outcome Data Generating Process

Outcomes follow a **Spatial Durbin Model (SDM)**:

$$\mathbf{Y} = (\mathbf{I} - \rho \mathbf{W})^{-1} [\tau \mathbf{Z} + \gamma \cdot \text{Spill}(\mathbf{Z}) + \beta \mathbf{X} + \varepsilon]$$

Parameter definitions:

Symbol	Role	Value / Range
τ	Direct treatment effect (estimand)	$\{0.8, 1.0, 1.5, 2.0, 3.0\}$; primary: 1.0
ρ	Outcome spatial autocorrelation	$\{0.00, 0.01, 0.20, 0.50\}$
γ	Spillover magnitude	$\{0.5, 0.6, 0.7, 0.8\}$
β	Incidence effect on outcome	1.0 (fixed)
σ	Residual standard deviation	1.0 (fixed)

Spillover construction:

$$\text{Spill}_i(\mathbf{Z}) = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|}$$

Two spillover modes are evaluated: - **control_only**: γ applied only to units with $Z_i = 0$ - **both**: γ applied to all units regardless of treatment status

Stationarity requires $|\rho| < 1/\lambda_{\max}(\mathbf{W})$; all swept ρ values satisfy this condition.

2.3 Incidence Generation

Three modes generate the cluster-level covariate $X_i \in [0, 1]$, each representing a different assumption about how historical incidence arises:

Mode 1 — iid Uniform:

$$X_i \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$$

No spatial structure; serves as the baseline. One config ($\rho_X = N/A$).

Mode 2 — Spatial (SAR filter):

$$\mathbf{X}^* = (\mathbf{I} - \rho_X \mathbf{W})^{-1} \mathbf{u}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$X_i = \Phi(X_i^*) \in (0, 1)$$

Spatially correlated via SAR filter; probability integral transform (Φ) ensures X_i is marginally $\text{Uniform}(0, 1)$ but spatially dependent. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Mode 3 — Poisson (count rates):

1. Spatially correlated log-rates: $\log \lambda_i^* = \mu_0 + [(\mathbf{I} - \rho_X \mathbf{W})^{-1} \mathbf{u}]_i$
2. Poisson counts: $C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$ where $P_i = 100,000$ (equal population per cluster; $\lambda_i^* P_i \approx 35$ expected deaths)
3. Observed rates: $R_i = C_i / P_i$
4. Rank-normalized: $X_i = (\text{rank}(R_i) - 0.5) / N$

Base rate $\mu_0 = \log(35/100,000)$ mimics SUD incidence. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Critical design invariant: Incidence surfaces $\mathbf{X}_1, \dots, \mathbf{X}_{10}$ are generated **once per (mode, ρ_X) configuration** and shared by all (ρ , γ , nb_type, spillover type) scenarios of that configuration; spatial model parameters affect the outcome DGP, not the incidence itself. **Matched surfaces:** for each surface k , treatment is assigned from $\mathbf{X}_k$, and the outcome and the analysis use the same $\mathbf{X}_k$. Reported metrics are therefore averages over incidence surfaces.

2.4 Treatment Assignment Designs

Eight assignment strategies are evaluated, representing qualitatively distinct approaches to allocating clusters to treatment (all treat exactly 50 of 100 except Isolation Buffer):

ID	Name	Deterministic?	Key Feature
1	Checkerboard	Yes	Alternating grid; maximizes spatial separation
2	High Incidence Focus	No	Treats the top-50 incidence clusters (random tie-breaking)
3	Saturation Quadrants	No	Random permutation of saturations $\{0.2, 0.4, 0.6, 0.8\}$ over quadrants

ID	Name	Deterministic?	Key Feature
4	Isolation Buffer	No	Greedy random maximal independent set; no adjacent treated clusters (mean treated: rook 38.2, queen 21.4)
5	2x2 Blocking	No	2 of 4 treated within each 2x2 spatial block
6	Balanced Quartiles	No	Stratified by incidence quartile; exactly 50 treated
7	Balanced Halves	No	Median split; 25 treated per half
8	Incidence-Guided Saturation Quadrants	No	Quadrant saturations {0.8, 0.6, 0.4, 0.2} by rank of quadrant mean incidence

Ties in incidence are broken at random, freshly for every design draw. Checkerboard is the only **deterministic** design: its assignment is identical on every draw, but each fit still receives its own outcome noise.

2.5 Estimation Methods

Difference in Means (DIM):

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1} \sum_{Z_i=1} Y_i - \frac{1}{n_0} \sum_{Z_i=0} Y_i$$

95% CI via Neyman variance estimator:

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$$

Maximum Likelihood (MLE via SAR):

Fits the spatial lag model $Y \sim Z + \text{Spill} + X$ by maximum likelihood with `fit_sar_lag()` (`04_estimation.R`), which matches `spatialreg::lagsarlm(method = "eigen")` to about 10^{-7} , simultaneously estimating ρ , τ , and regression coefficients. The true spillover covariate $\text{Spill}(\mathbf{Z})$ is included as an **oracle regressor** (i.e., the researcher is assumed to know the true spillover structure). 95% Wald CIs use the ML standard error. A non-oracle fit ($Y \sim Z + X$) was run as a sensitivity analysis; it is not shown in this report. Under rook contiguity, Checkerboard has $W\mathbf{Z} = \mathbf{1} - \mathbf{Z}$, so the oracle Spill column is aliased and τ is not identified; those estimates are kept but flagged.

3 Simulation Algorithm

3.1 Parameter Grid

Table 1: Simulation parameter grid. Primary tau=1.0 slice: 2,560 scenarios. Full tau-sweep (tau in 0.8, 1.0, 1.5, 2.0, 3.0): 12,800 total scenarios.

Parameter	Values	Levels
Incidence mode	iid, spatial, poisson	5 configs (1 iid + 2 spatial + 2 poisson)
rho_X (incidence spatial autocorr.)	N/A (iid); 0.20, 0.50 (spatial, poisson)	—
nb_type (contiguity)	rook, queen	2
rho (outcome spatial autocorr.)	0.00, 0.01, 0.20, 0.50	4
gamma (spillover magnitude)	0.5, 0.6, 0.7, 0.8	4
spill_type	control_only, both	2

design_id	1, 2, 3, 4, 5, 6, 7, 8	8
TOTAL		2,560 scenarios (primary tau=1.0 slice)

3.2 Resampling Structure

For each MLE scenario, the simulation uses $K = 10$ incidence surfaces and $J = 25$ design draws per surface; every one of the $K \cdot J = 250$ fits has its own outcome noise column. The DIM baseline (pre-revision) used a different scheme:

Mode	Design draws	Outcome draws	Fits / scenario
DIM (pre-revision)	25	100	2,500
MLE	25 per surface	10 surfaces	250

3.3 Evaluation Metrics

For each scenario, across the n valid fits $i = 1, \dots, n$:

$$\text{Bias} = \bar{\hat{\tau}} - \tau, \quad \text{SD} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\hat{\tau}_i - \bar{\hat{\tau}})^2}$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i - \tau)^2$$

$$\text{Coverage} = \frac{1}{n} \sum_{i=1}^n \mathbf{1} [\tau \in (\hat{\tau}_{i,L}, \hat{\tau}_{i,U})]$$

Monte Carlo standard errors are computed from the 10 surface-level means (the surface is the primary sampling unit), not from the 250 fits treated as independent.

3.4 Per-Draw Reproducibility

Every random draw is preceded by a key-based seed, `set.seed(digest::digest2int(key))`, where the key joins the fields that define the draw with “|”: (“X”, mode, ρ_X , k) for surface k ; (“Z”, mode, ρ_X , nb_type, ρ , γ , spill_type, k , d) for design draws; (“eps”, mode, ρ_X , nb_type, ρ , γ , spill_type) for outcome noise.

Each draw depends only on its key, so results are identical in any run order. Within a block, all designs share the same noise and incidence surfaces, and every τ level reuses the same assignments and noise (common random numbers).

4 Results: Difference in Means (DIM)

DIM is a naive baseline. All results in this section are **pre-revision** (March 2026 run: old data-generating process, 6 designs, no Balanced Halves or Incidence-Guided Saturation Quadrants); design labels use the current numbering. Tables pool rook and queen.

4.1 Overall Summary (All DIM Scenarios)

Table 2: DIM (pre-revision): Summary statistics across all 1920 scenarios

Metric	Min	Mean	Max
Bias	-0.883	-0.204	0.558
MSE	0.032	0.152	0.799
Coverage	0.000	0.720	0.990

4.2 DIM Design Comparison by MSE

Table 3: DIM (pre-revision): Design performance averaged across all scenarios and incidence modes

Design	Mean Bias	Median MSE	Mean MSE	Mean Coverage
Design 6	-0.128	0.091	0.095	0.821
Design 2	-0.076	0.081	0.101	0.822
Design 3	-0.011	0.090	0.111	0.802
Design 5	-0.253	0.101	0.128	0.736
Design 4	-0.310	0.133	0.163	0.702
Design 1	-0.448	0.253	0.314	0.446

4.3 DIM: MSE by Design and Neighbor Type

Distribution of Estimation Error (MSE) by Spatial Contiguity

Incidence: iid Uniform

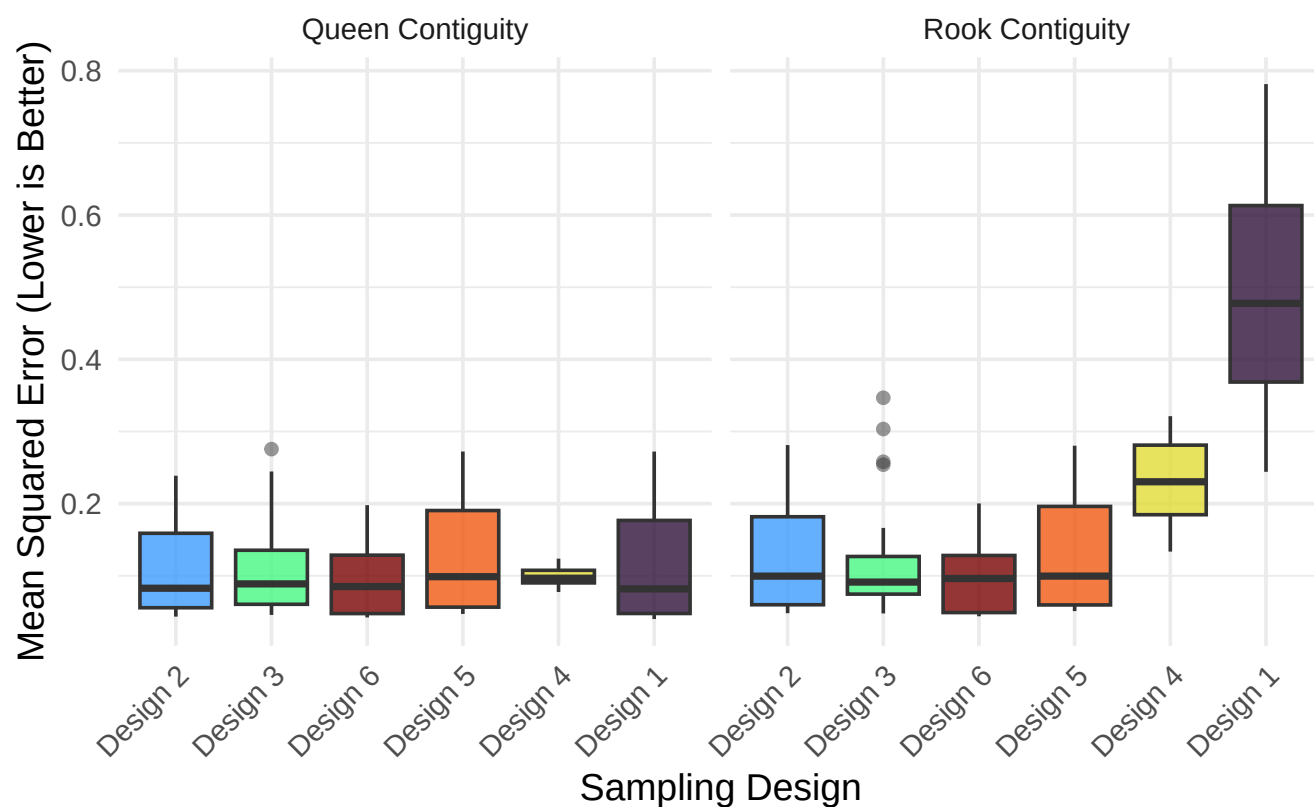


Figure 1: DIM (pre-revision): Distribution of MSE by design, faceted by spatial contiguity structure. Boxes sorted by median MSE within each panel.

4.4 DIM: MSE Heatmap (Rho x Design)

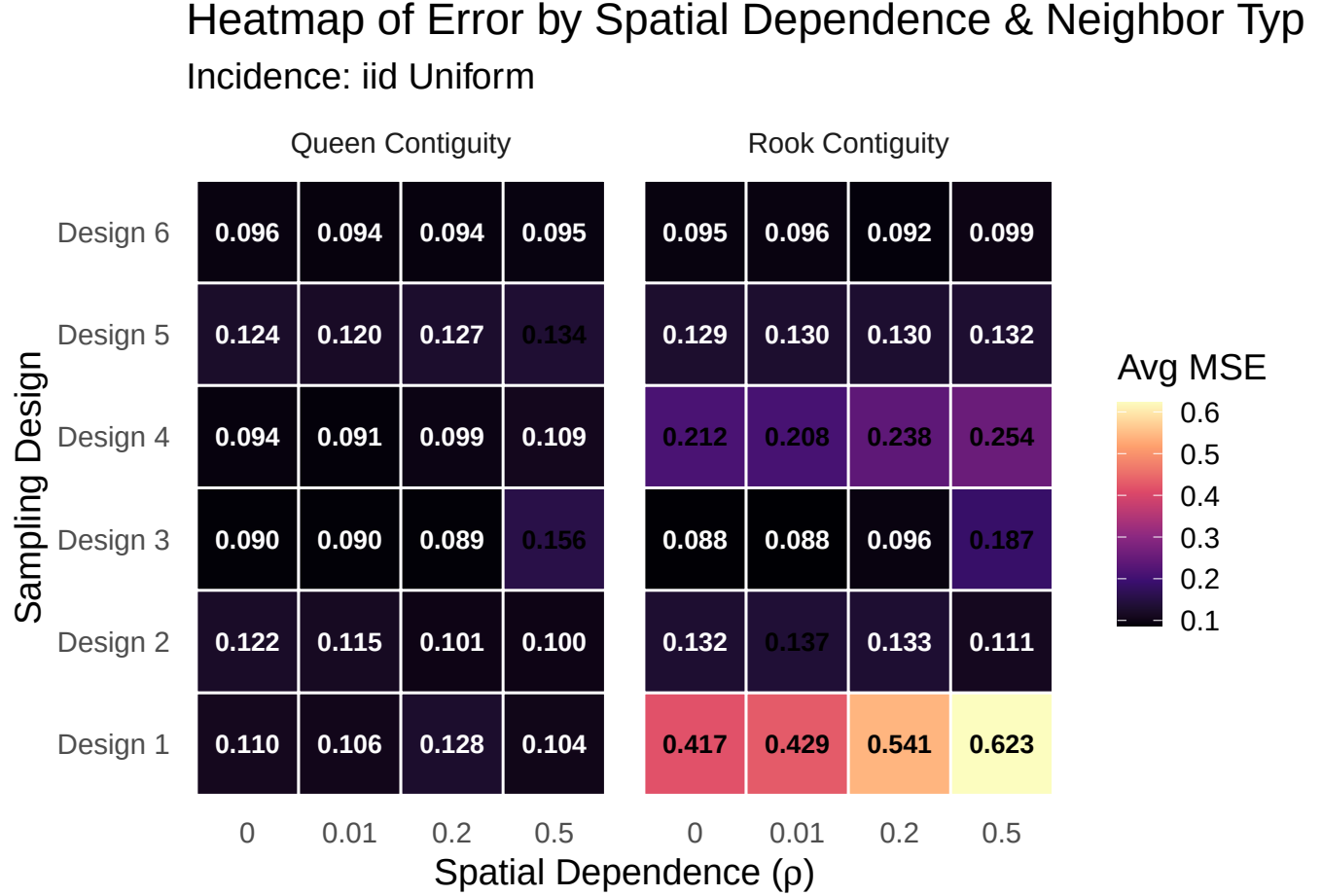


Figure 2: DIM (pre-revision) MSE heatmap for iid Uniform incidence. Cells show average MSE across spillover type and gamma, faceted by neighbor type.

4.5 DIM: Design Rank Frequency

Table 4: DIM (pre-revision): Design rank frequency (%) for iid Uniform incidence. Each cell = % of scenarios where design achieved that rank (Rank 1 = lowest MSE).

Design	Rank_1	Rank_2	Rank_3	Rank_4	Rank_5	Rank_6
Design 3	32.8	29.7	7.8	9.4	14.1	6.2
Design 6	28.1	34.4	31.2	6.2	0.0	0.0
Design 4	15.6	7.8	4.7	4.7	48.4	18.8
Design 2	12.5	4.7	21.9	39.1	20.3	1.6
Design 1	10.9	7.8	4.7	14.1	6.2	56.2
Design 5	0.0	15.6	29.7	26.6	10.9	17.2

4.6 DIM: Robustness Profile

Table 5: DIM (pre-revision): Robustness profile for iid Uniform incidence. Sorted by Median MSE. IQR = interquartile range across scenarios.

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0434	0.0574	0.0897	0.1754	0.2812	0.1180
Design 3	0.0458	0.0605	0.0904	0.1306	0.3467	0.0701
Design 6	0.0423	0.0478	0.0945	0.1284	0.2003	0.0806
Design 5	0.0472	0.0575	0.0996	0.1933	0.2803	0.1358
Design 4	0.0775	0.0969	0.1287	0.2295	0.3213	0.1326
Design 1	0.0401	0.0962	0.2441	0.4764	0.7815	0.3801

4.7 DIM: Coverage by Design

95% CI Coverage Probability by Design

Incidence: iid Uniform | Red dashed = nominal 95%

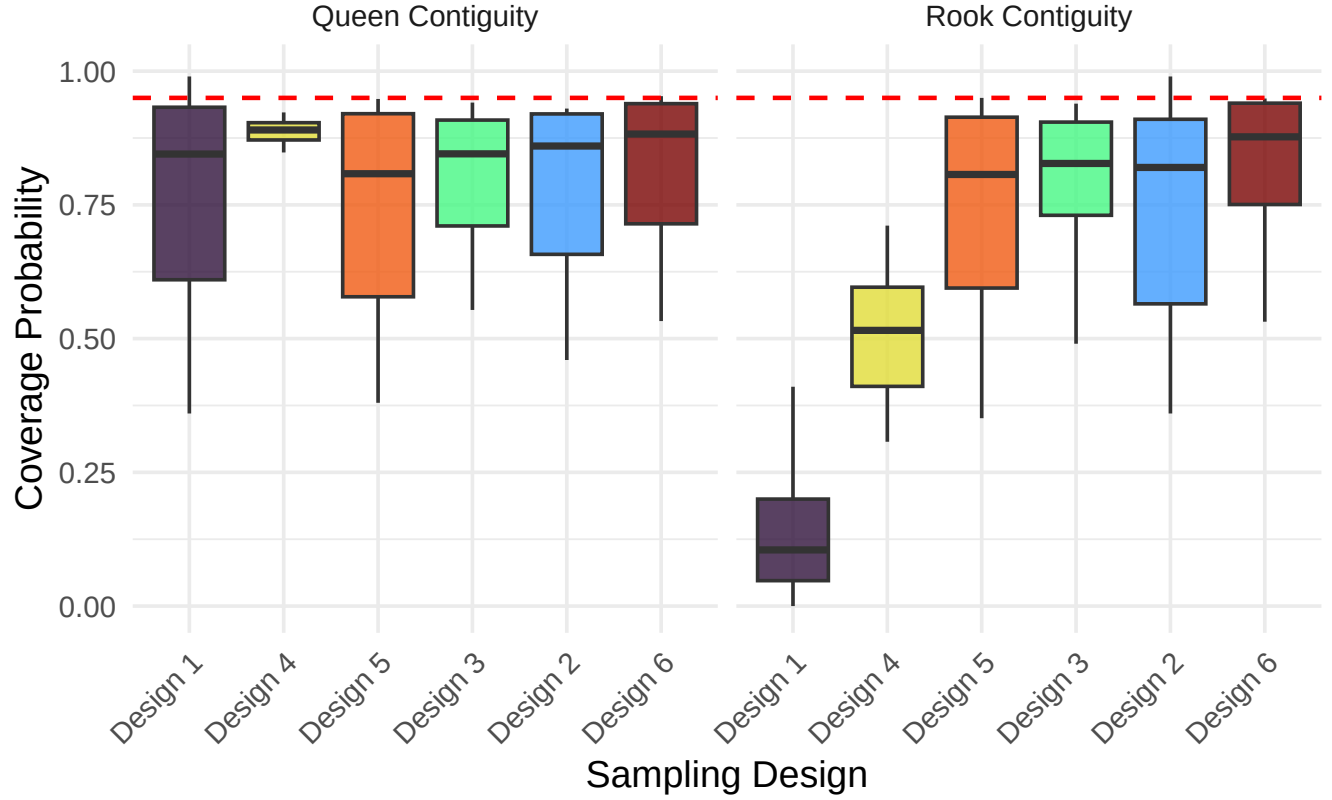


Figure 3: DIM (pre-revision): Coverage probability distribution by design across all scenarios. Red dashed line = nominal 95% target.

4.8 DIM: Bias-Variance Decomposition

Bias-Variance Decomposition of MSE

Incidence: iid Uniform

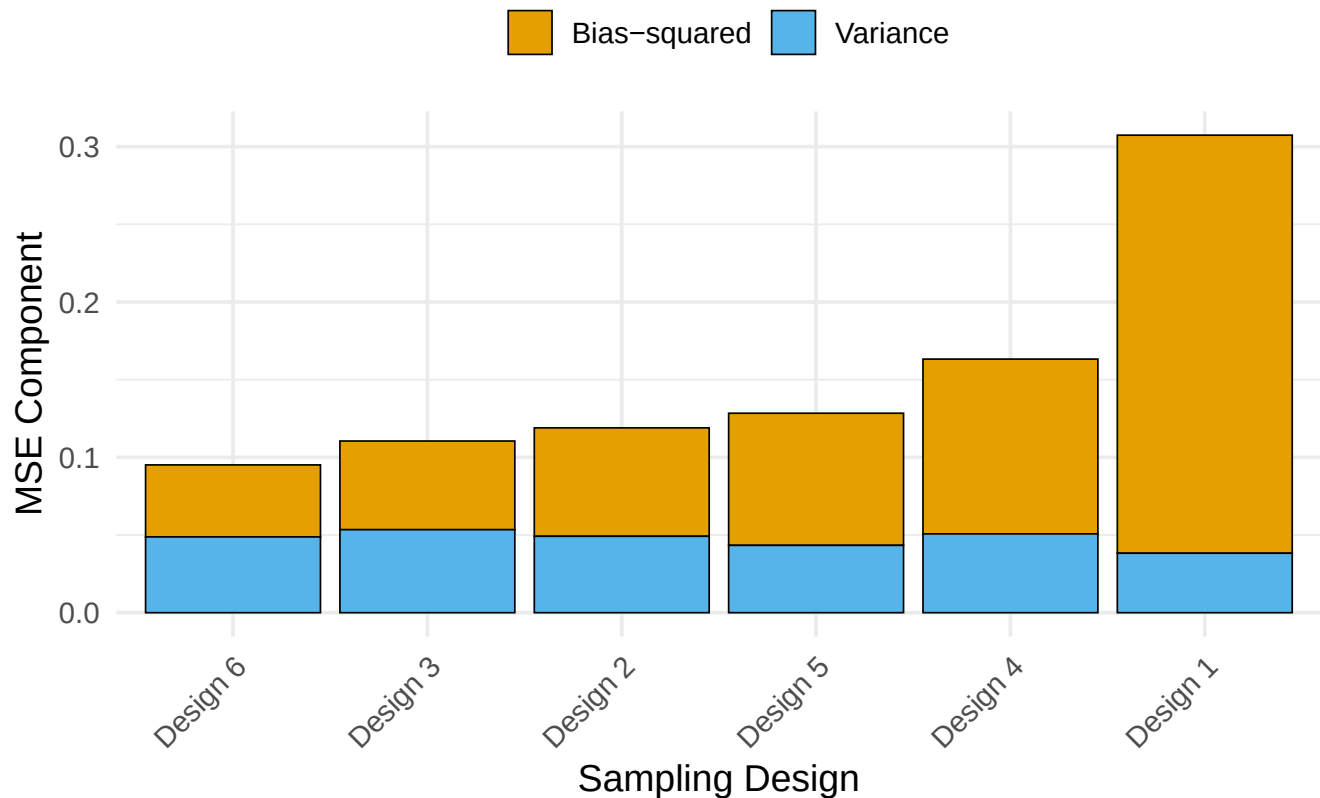


Figure 4: DIM (pre-revision): Bias-squared (dark) and variance (light) components of MSE per design for iid Uniform incidence. Total bar height = MSE.

5 Results: Maximum Likelihood (MLE)

5.1 Overall Summary (All MLE Scenarios)

Table 6: MLE: Summary statistics across all 2560 $\tau = 1.0$ scenarios, pooled over rook and queen and over incidence modes. The minimum coverage comes from Checkerboard under rook, where τ is not identified.

Metric	Min	Mean	Max
Bias	-0.827	-0.028	0.260
MSE	0.032	0.219	2.843
Coverage	0.000	0.890	0.990

Fail Rate	NA	0.000	NA
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5.2 MLE Design Comparison by MSE

Table 7: MLE: Design performance at $\tau = 1.0$ by neighbor type (queen primary, rook sensitivity), averaged across incidence modes. Checkerboard under rook: τ not identified ($WZ = 1 - Z$); estimates kept but flagged.

Neighbors	Design	Mean Bias	Median MSE	Mean MSE	Mean Coverage	Fail Rate
queen	Design 3	0.025	0.079	0.090	0.936	0
queen	Design 8	0.032	0.080	0.091	0.937	0
queen	Design 7	0.018	0.114	0.130	0.936	0
queen	Design 6	0.017	0.115	0.131	0.936	0
queen	Design 4	0.003	0.159	0.160	0.939	0
queen	Design 2	0.031	0.234	0.245	0.938	0
queen	Design 5	0.010	0.271	0.319	0.939	0
queen	Design 1	0.001	0.899	1.087	0.941	0
rook	Design 3	0.011	0.065	0.071	0.940	0
rook	Design 8	0.018	0.069	0.072	0.942	0
rook	Design 6	0.008	0.078	0.084	0.940	0
rook	Design 7	0.008	0.079	0.084	0.940	0
rook	Design 5	0.004	0.117	0.128	0.938	0
rook	Design 4	0.003	0.133	0.133	0.939	0
rook	Design 2	0.018	0.194	0.206	0.939	0
rook	Design 1	-0.652	0.470	0.480	0.150	0

5.3 MLE: MSE by Design and Neighbor Type

Distribution of Estimation Error (MSE) by Spatial Contiguity

Incidence: iid Uniform

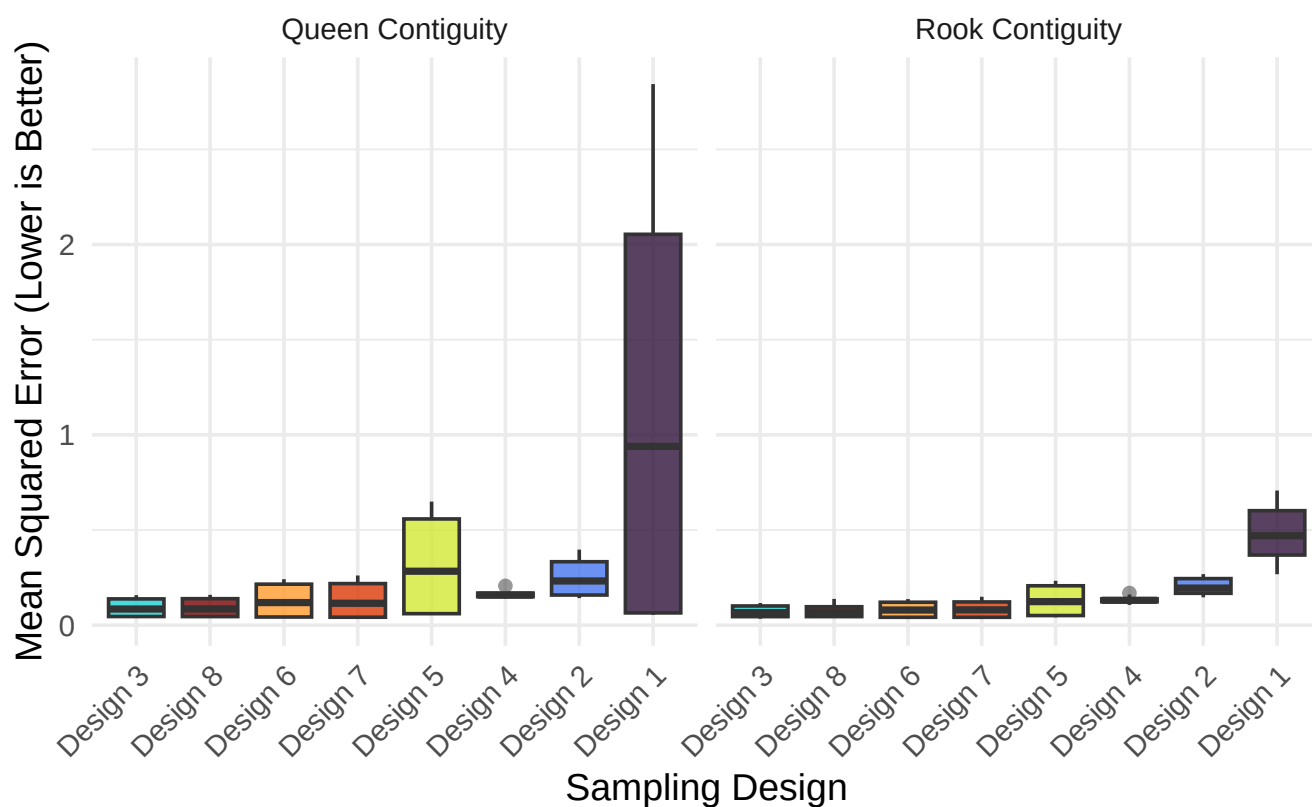


Figure 5: MLE: Distribution of MSE by design at $\tau = 1.0$ (iid Uniform incidence), faceted by spatial contiguity (queen primary, rook sensitivity). Checkerboard under rook: τ not identified; estimates kept but flagged.

5.4 MLE: MSE Heatmap

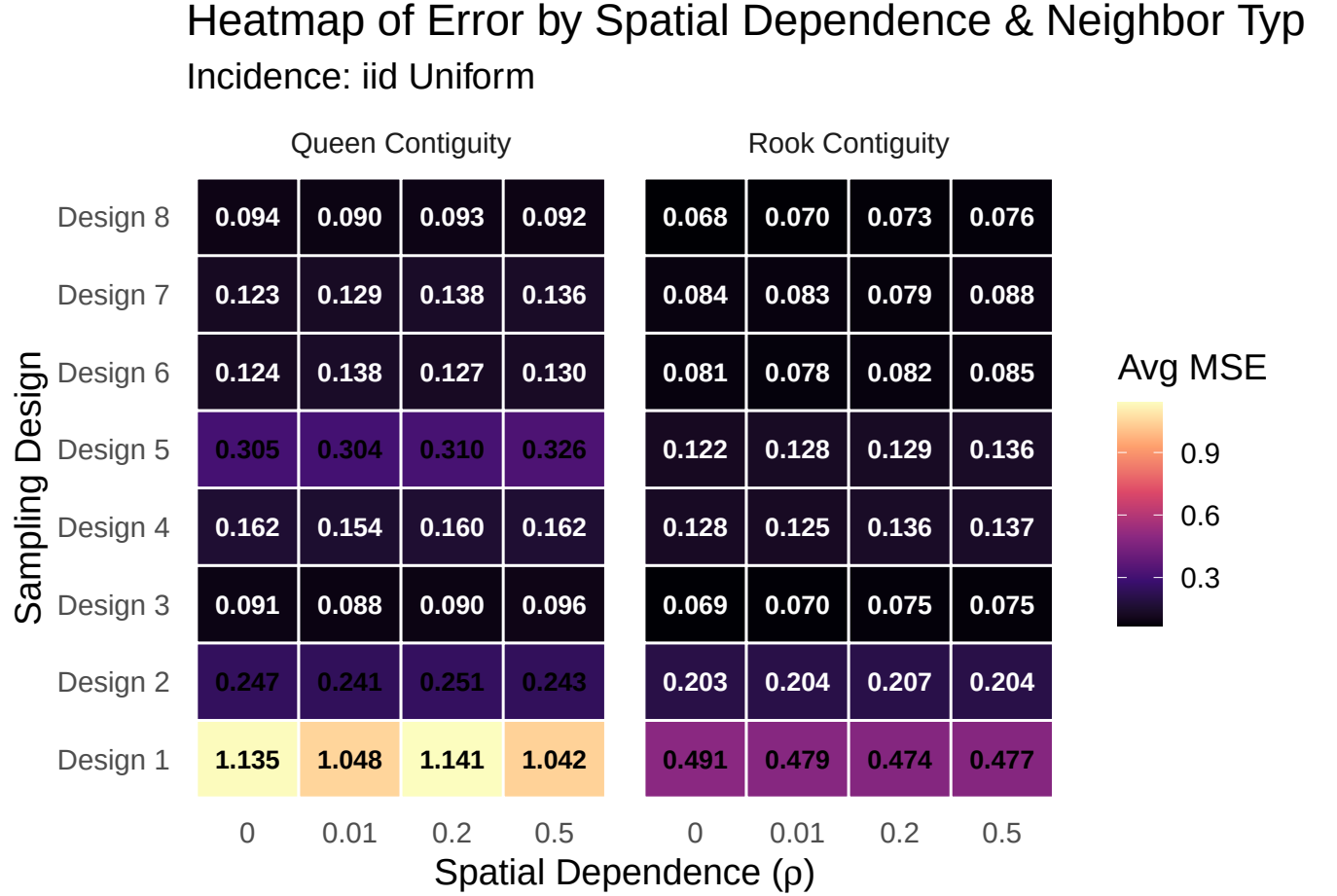


Figure 6: MLE MSE heatmap for iid Uniform incidence at $\tau = 1.0$, faceted by neighbor type. Cells show average MSE across spillover type and gamma.

5.5 MLE: Design Rank Frequency

Table 8: MLE: Design rank frequency (%) for iid Uniform incidence, queen contiguity, $\tau = 1.0$.

Design	Rank_5	Rank_6	Rank_8	Rank_7	Rank_1	Rank_2	Rank_3	Rank_4
Design 3	0.0	0.0	0	0	40.6	25.0	18.8	15.6
Design 8	0.0	0.0	0	0	21.9	31.2	25.0	21.9
Design 7	25.0	0.0	0	0	18.8	15.6	12.5	28.1
Design 6	25.0	0.0	0	0	12.5	25.0	3.1	34.4
Design 4	0.0	0.0	25	25	6.2	3.1	40.6	0.0
Design 1	21.9	28.1	50	0	0.0	0.0	0.0	0.0
Design 2	0.0	50.0	25	25	0.0	0.0	0.0	0.0
Design 5	28.1	21.9	0	50	0.0	0.0	0.0	0.0

5.6 MLE: Robustness Profile

Table 9: MLE: Robustness profile for iid Uniform incidence, queen contiguity, $\tau = 1.0$.

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0377	0.0454	0.0848	0.1381	0.1576	0.0927
Design 8	0.0392	0.0451	0.0853	0.1392	0.1594	0.0941
Design 7	0.0373	0.0417	0.1150	0.2185	0.2610	0.1767
Design 6	0.0381	0.0428	0.1187	0.2153	0.2416	0.1725
Design 4	0.1410	0.1495	0.1580	0.1686	0.2065	0.0191
Design 2	0.1423	0.1584	0.2323	0.3334	0.3970	0.1750
Design 5	0.0544	0.0604	0.2835	0.5581	0.6493	0.4977
Design 1	0.0547	0.0645	0.9394	2.0536	2.8430	1.9892

5.7 MLE: Coverage by Design

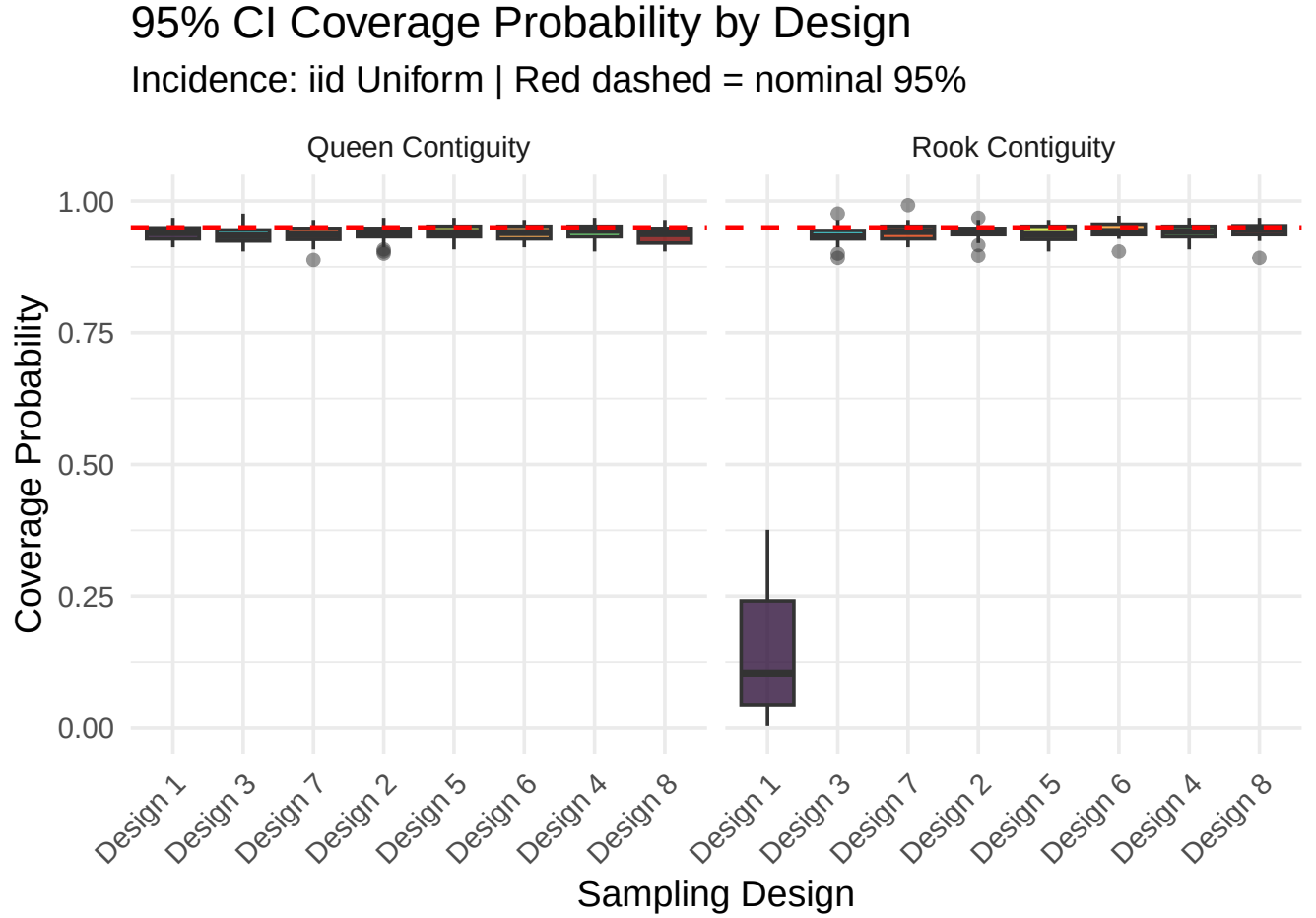


Figure 7: MLE: Coverage probability distribution by design (iid Uniform incidence, $\tau = 1.0$), faceted by neighbor type. Coverage is near-nominal for every design except Checkerboard under rook, where τ is not identified. Red dashed line = 95%.

6 DIM vs. MLE Comparison

6.1 Side-by-Side Metrics

Table 10: DIM vs. MLE: Average MSE and coverage by design, pooled over rook and queen and over incidence modes (MLE at $\tau = 1.0$). DIM is the pre-revision baseline (old data-generating process, 6 designs), so the two columns are not like-for-like; Design 7 (Balanced Halves) and Design 8 (Incidence-Guided Saturation Quadrants) were not in the DIM run and are NA by construction. Pooled MLE values for Checkerboard mix queen with rook, where τ is not identified.

Design	DIM (pre-revision)		MLE	
	DIM MSE	DIM Coverage	MLE MSE	MLE Coverage
Design 3	0.111	0.802	0.080	0.938
Design 8	NA	NA	0.081	0.939
Design 6	0.095	0.821	0.107	0.938
Design 7	NA	NA	0.107	0.938
Design 4	0.163	0.702	0.147	0.939
Design 5	0.128	0.736	0.224	0.938
Design 2	0.101	0.822	0.226	0.938
Design 1	0.314	0.446	0.783	0.545

6.2 Coverage Comparison by Estimator

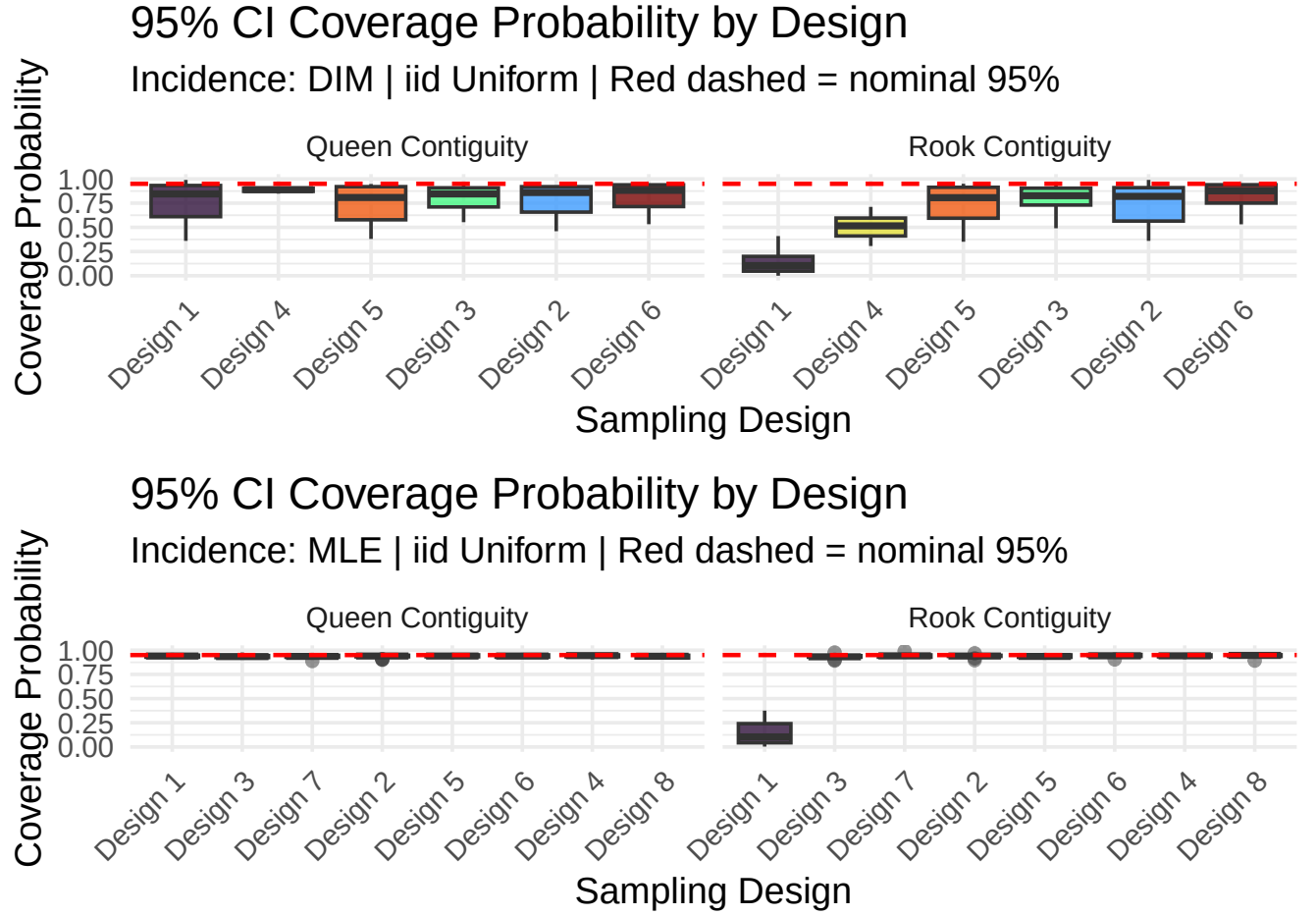


Figure 8: Coverage comparison: DIM (top, pre-revision baseline) vs MLE (bottom, $\tau = 1.0$) for the iid Uniform incidence configuration. MLE achieves near-nominal coverage for every design except Checkerboard under rook; DIM coverage is mostly below 0.95.

6.3 Key Observations

- **Coverage does not separate the designs.** MLE coverage averages 0.936–0.942 for every design and neighbor type except Checkerboard under rook, i.e. near-nominal (a finite-sample Wald shortfall of about one point). MSE is what separates the designs.
- **DIM baseline (pre-revision):** mean coverage 0.72 across its 1920 scenarios. It was run under the old data-generating process, so it is not a like-for-like comparison with the MLE results.
- **Checkerboard (Design 1):** highest MSE under queen (1.087 at $\tau = 1$), with mean bias 0.001, so almost all of it is variance: the oracle model separates τ from the spillover term only weakly under a checkerboard. Under rook, $WZ = \mathbf{1} - Z$, so τ is not identified: bias -0.652 (approximately $-\gamma$), coverage 0.150. These estimates are kept but flagged.
- **Lowest MSE (queen):** Saturation Quadrants (0.090) and Incidence-Guided Saturation Quadrants (0.091), which are statistically indistinguishable (eight-design comparisons in `results/eight_design_supplementary/`)

7 Sensitivity Analysis

7.1 Parameter Sensitivity (DIM)

Table 11: DIM (pre-revision): Eta-squared (proportion of MSE variance explained) by parameter, for iid Uniform incidence. Higher values indicate parameters where design choice matters more.

Design	Rho	Gamma	Spillover_Type	Neighbor_Type
Design 1	0.0350	0.1088	0.0202	0.7220
Design 2	0.0147	0.1018	0.7598	0.0172
Design 3	0.2809	0.0582	0.0005	0.0042
Design 4	0.0274	0.1661	0.0001	0.7196
Design 5	0.0015	0.1172	0.7968	0.0006
Design 6	0.0011	0.1200	0.6661	0.0001

7.2 Effect of Rho on Design Rankings (DIM)

MSE Dynamics for Design 1

Incidence: iid Uniform

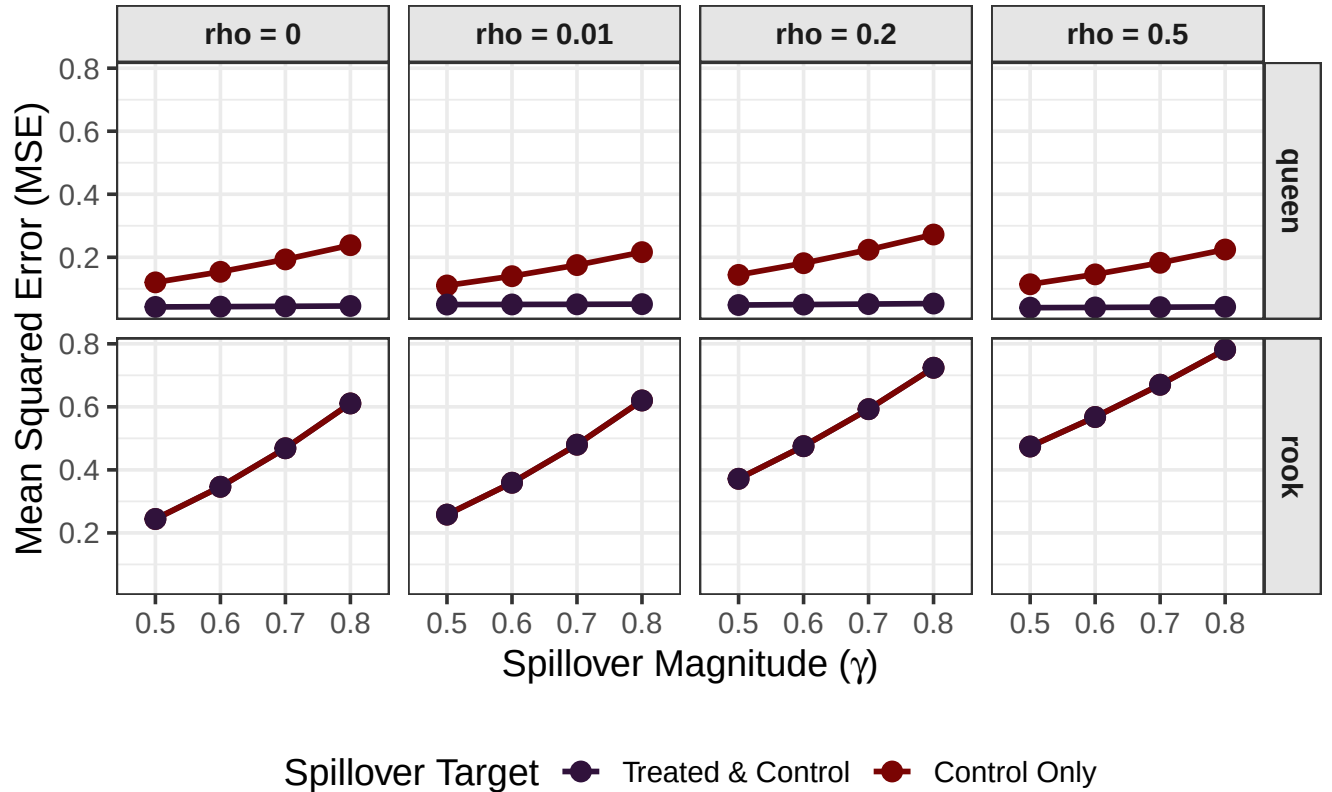


Figure 9: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

MSE Dynamics for Design 2

Incidence: iid Uniform

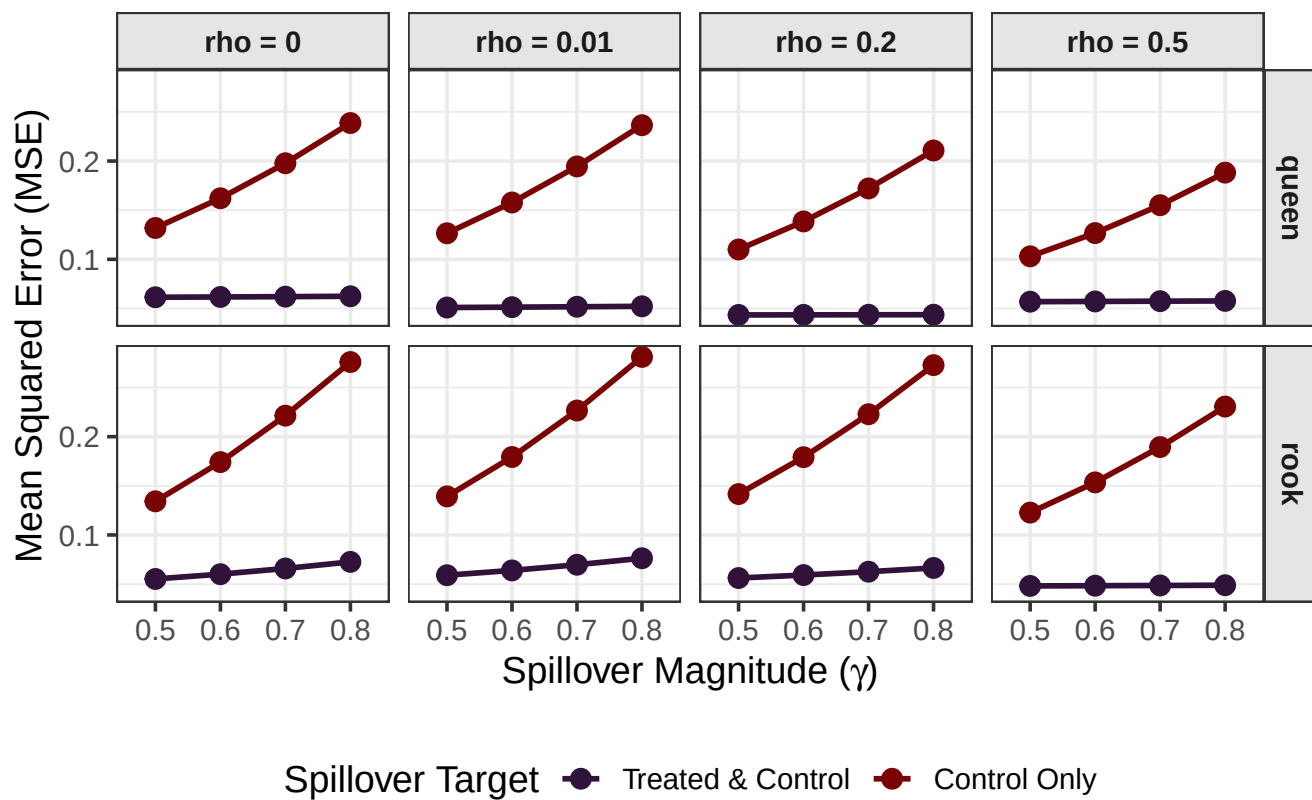


Figure 10: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

MSE Dynamics for Design 3

Incidence: iid Uniform

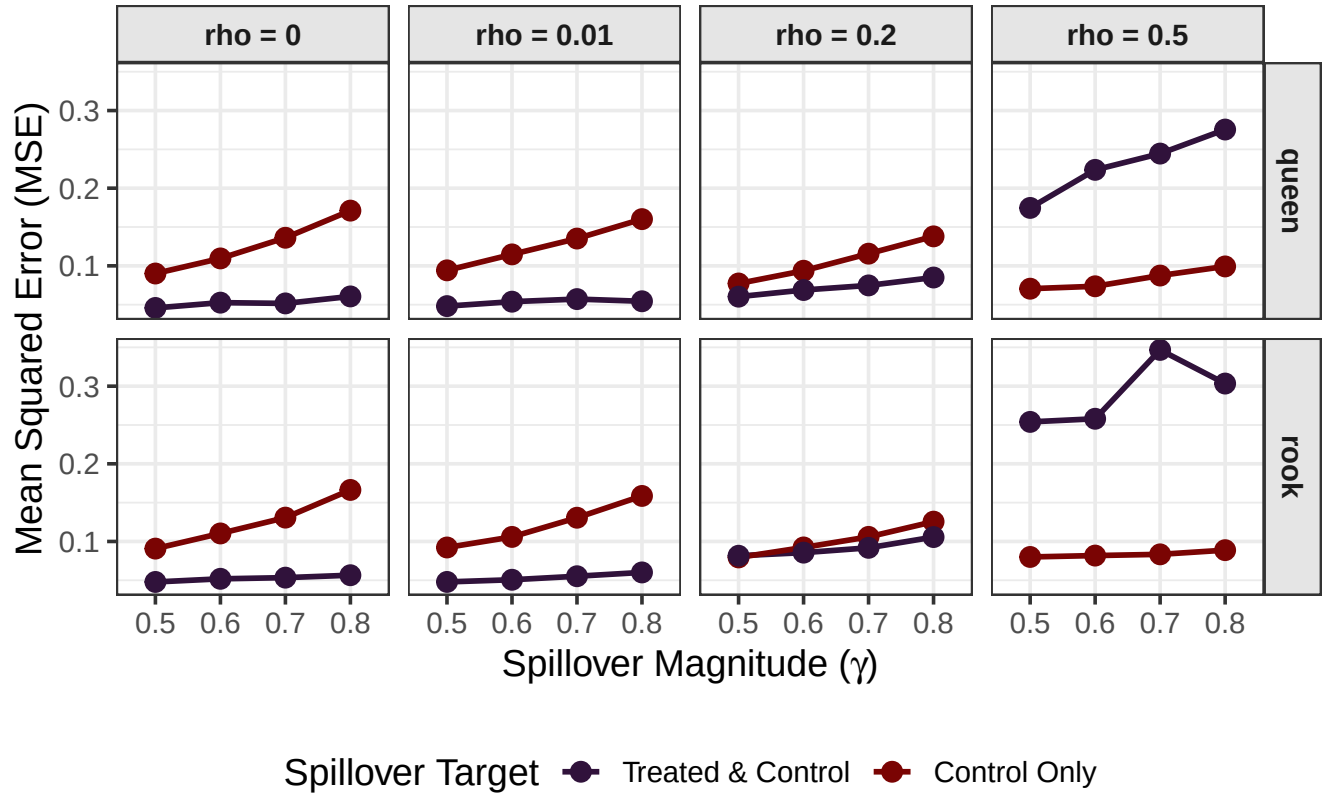


Figure 11: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

MSE Dynamics for Design 4

Incidence: iid Uniform

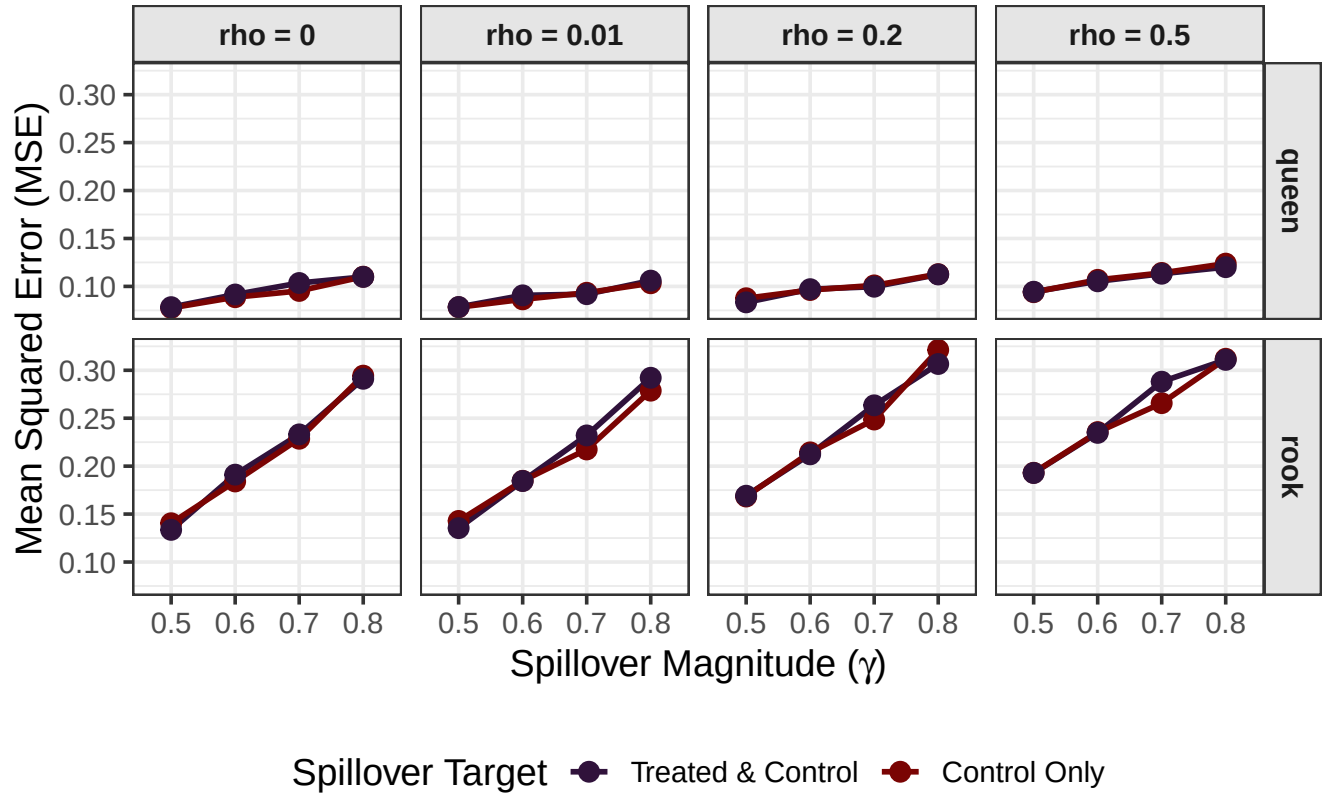


Figure 12: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

MSE Dynamics for Design 5

Incidence: iid Uniform

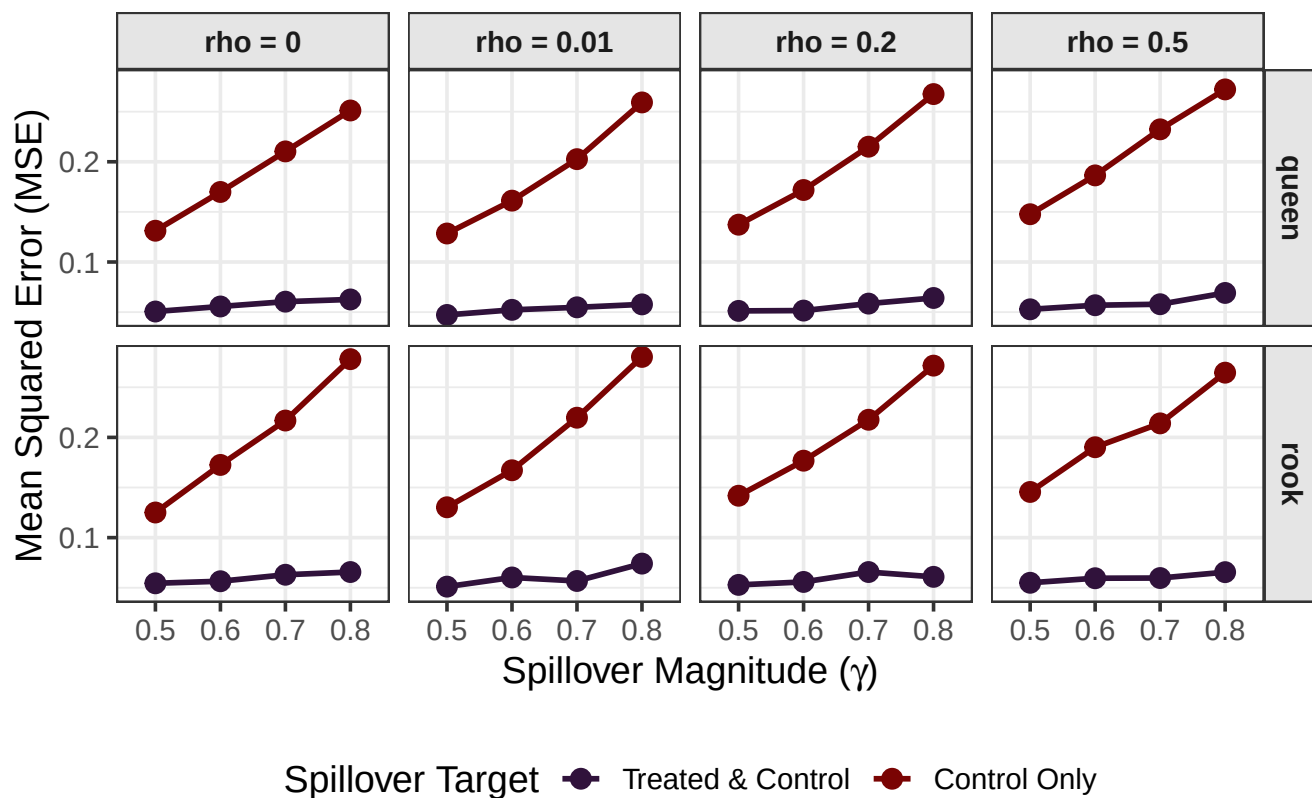


Figure 13: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

MSE Dynamics for Design 6

Incidence: iid Uniform

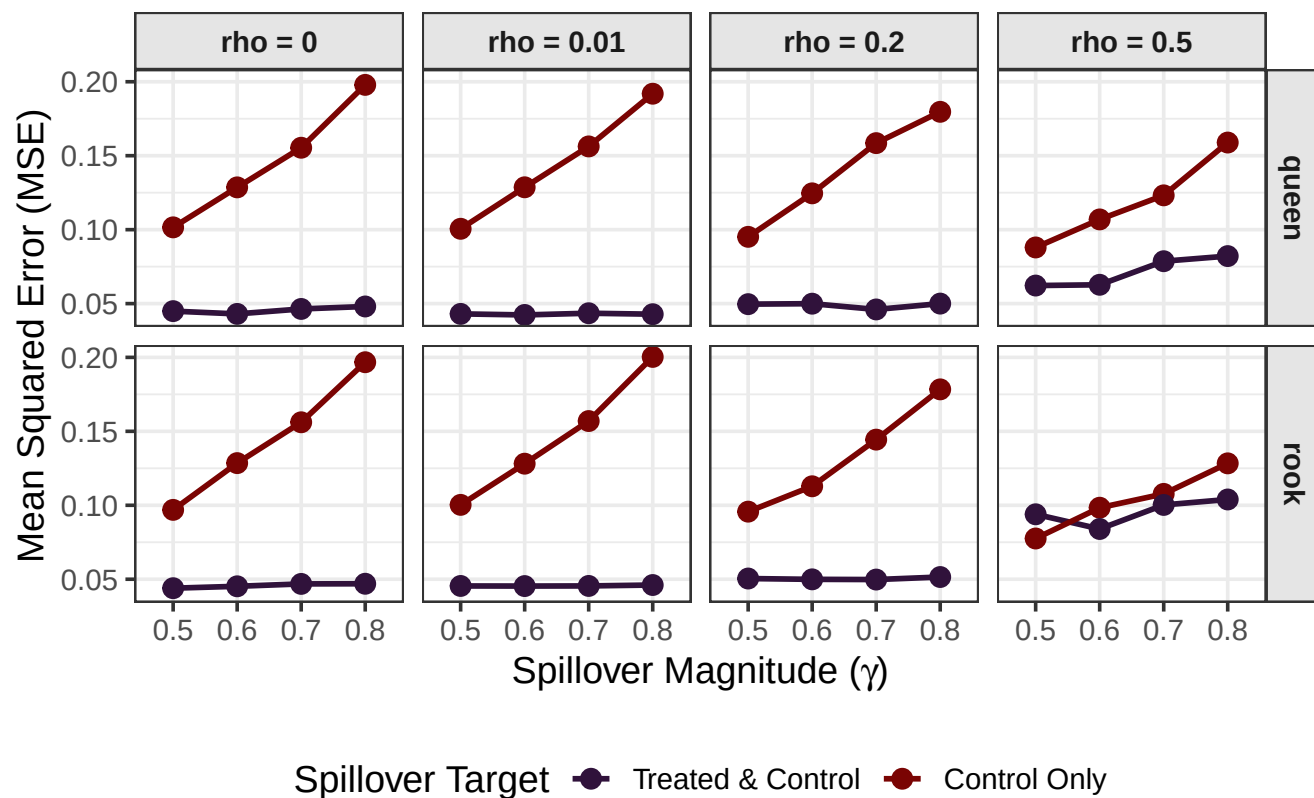


Figure 14: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

\$`Design 1`

MSE Dynamics for Design 1

Incidence: iid Uniform

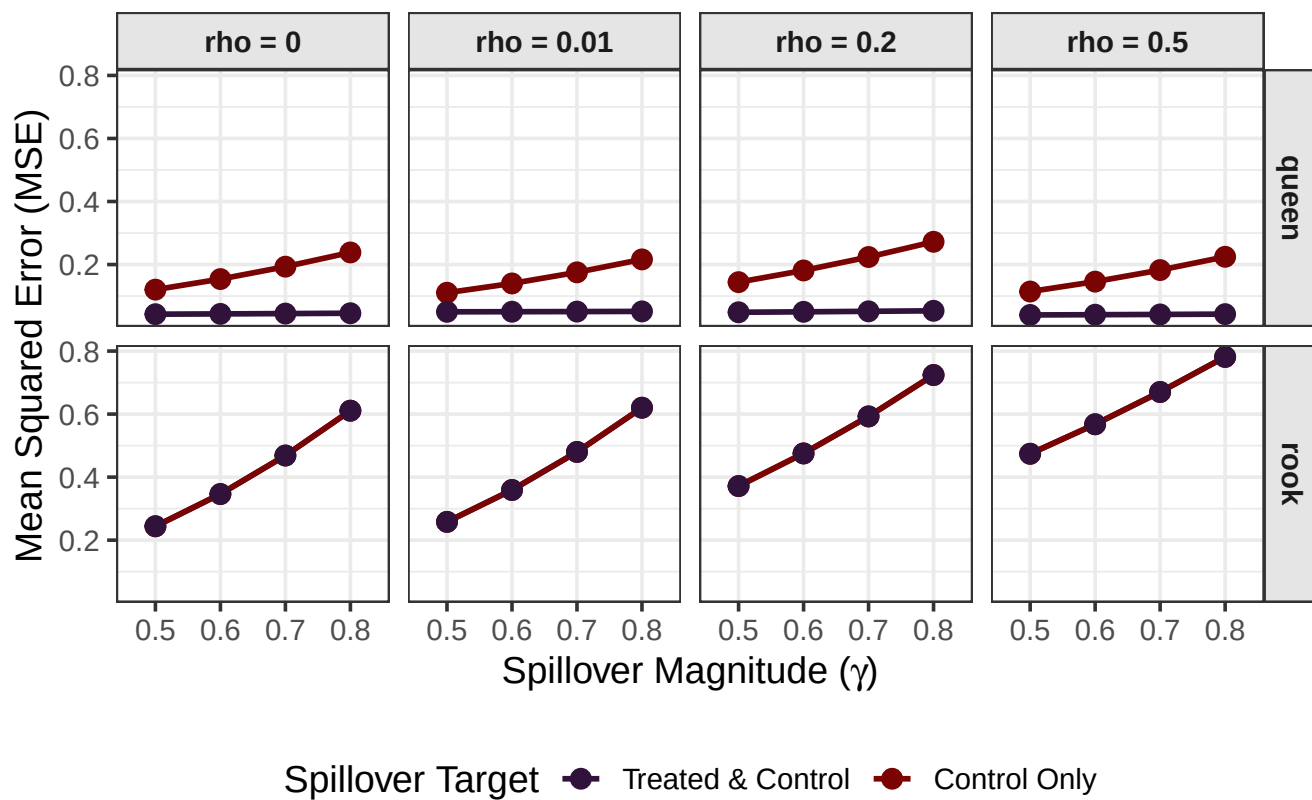


Figure 15: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 2`
```

MSE Dynamics for Design 2

Incidence: iid Uniform

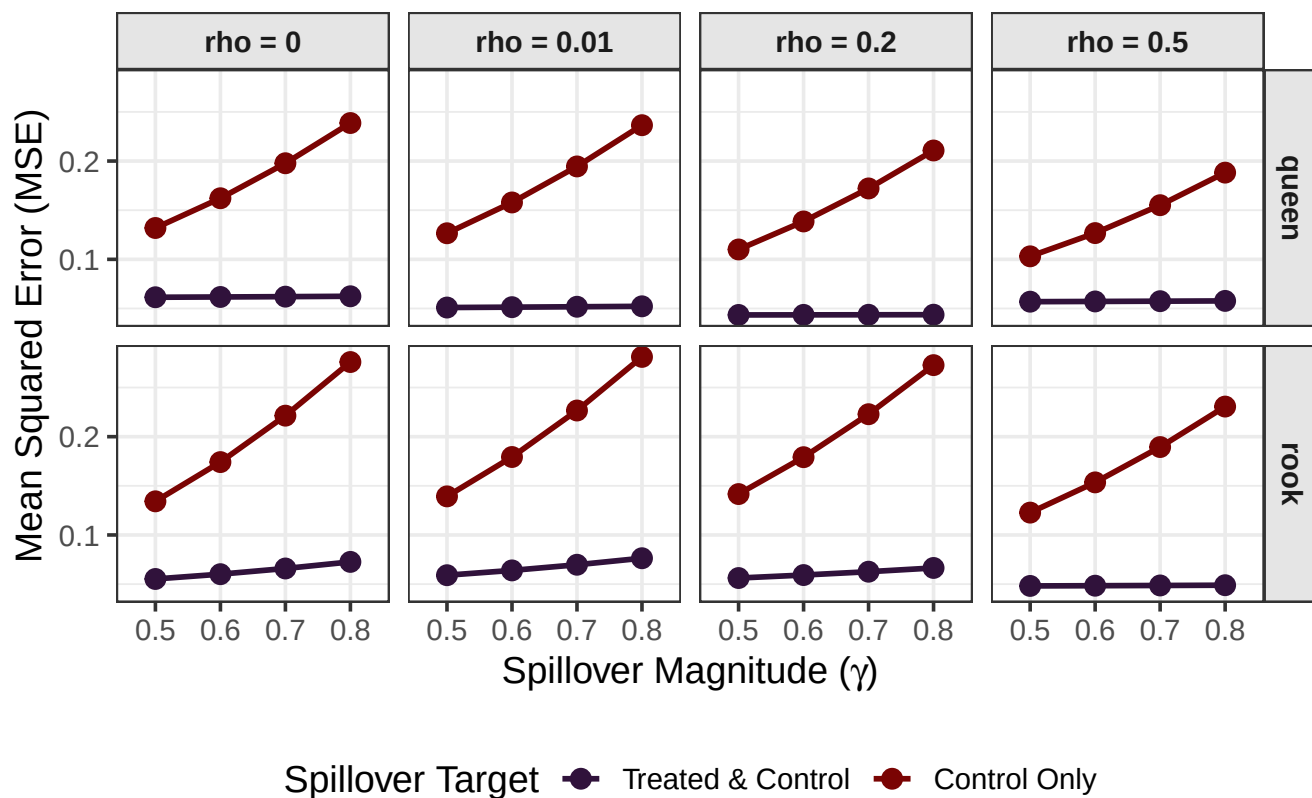


Figure 16: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 3`
```

MSE Dynamics for Design 3

Incidence: iid Uniform

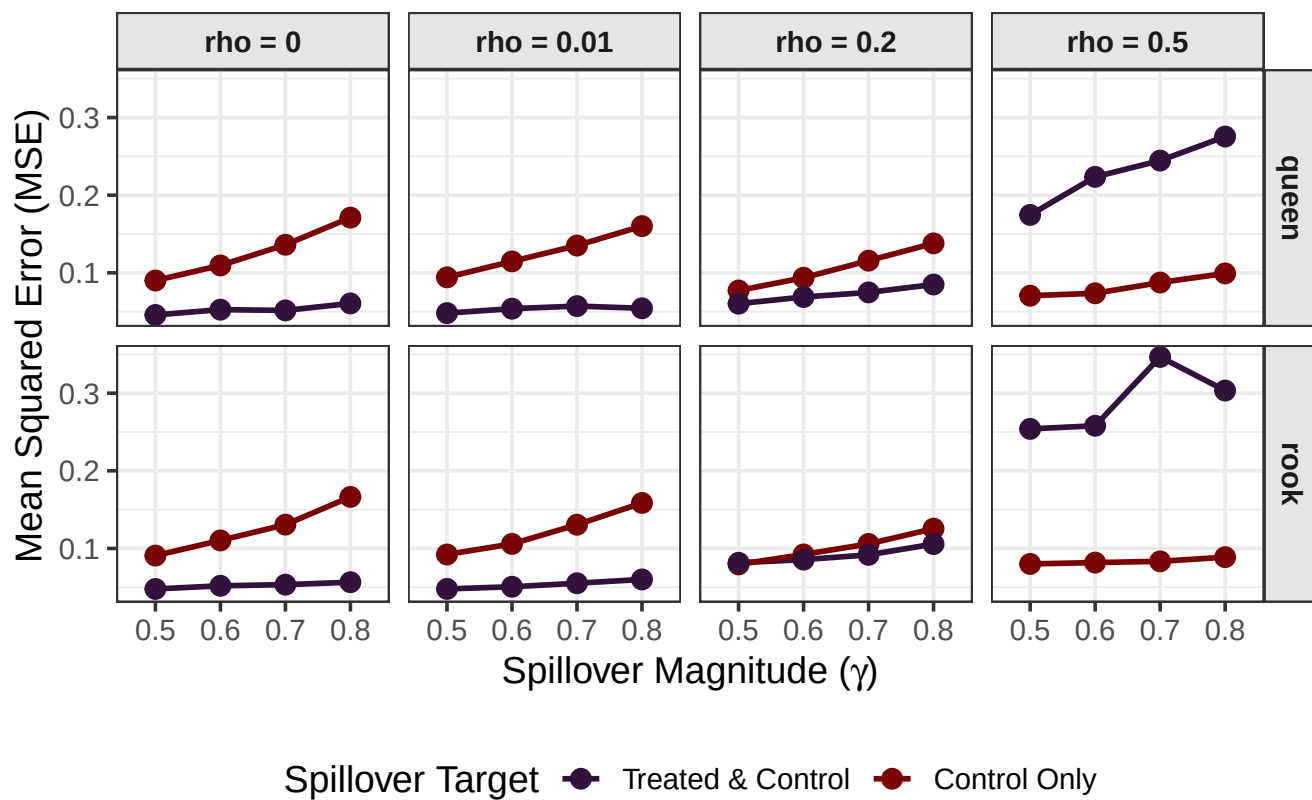


Figure 17: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 4`
```

MSE Dynamics for Design 4

Incidence: iid Uniform

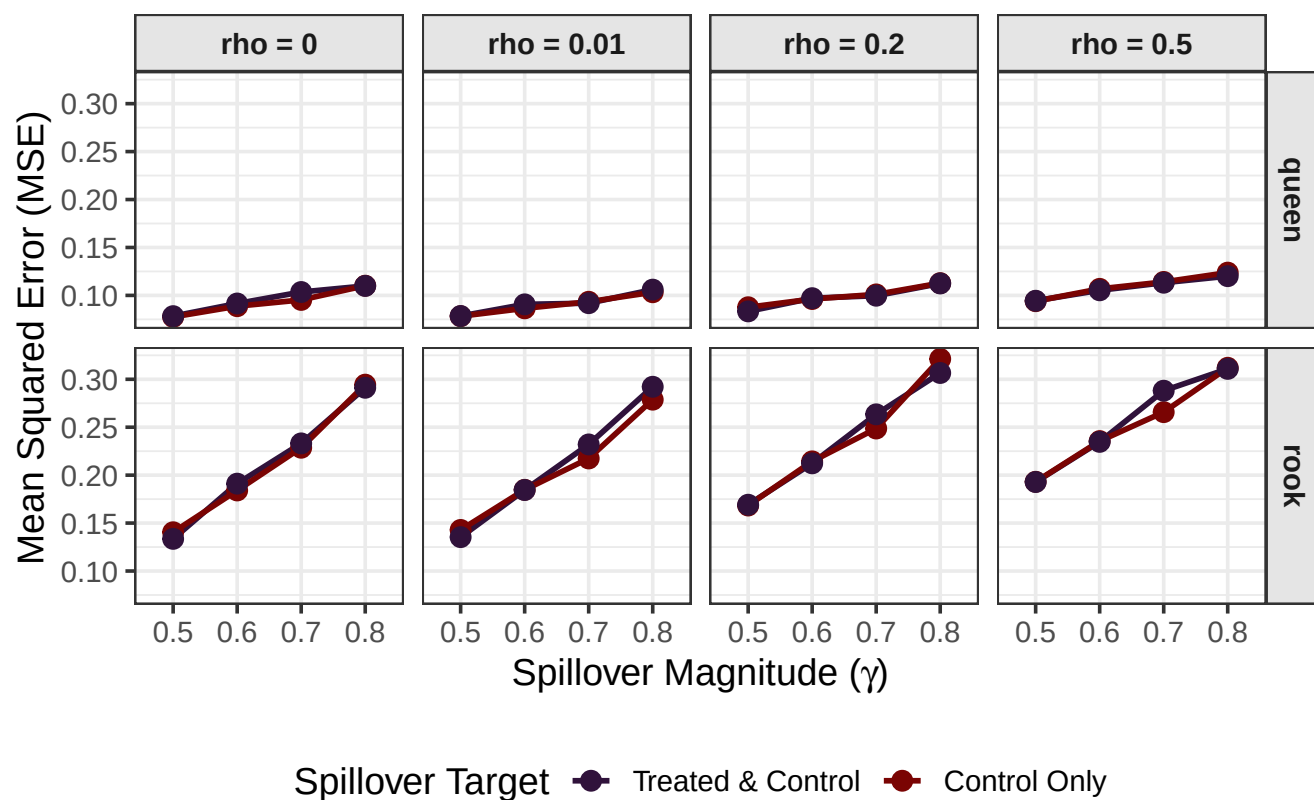


Figure 18: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 5`
```

MSE Dynamics for Design 5

Incidence: iid Uniform

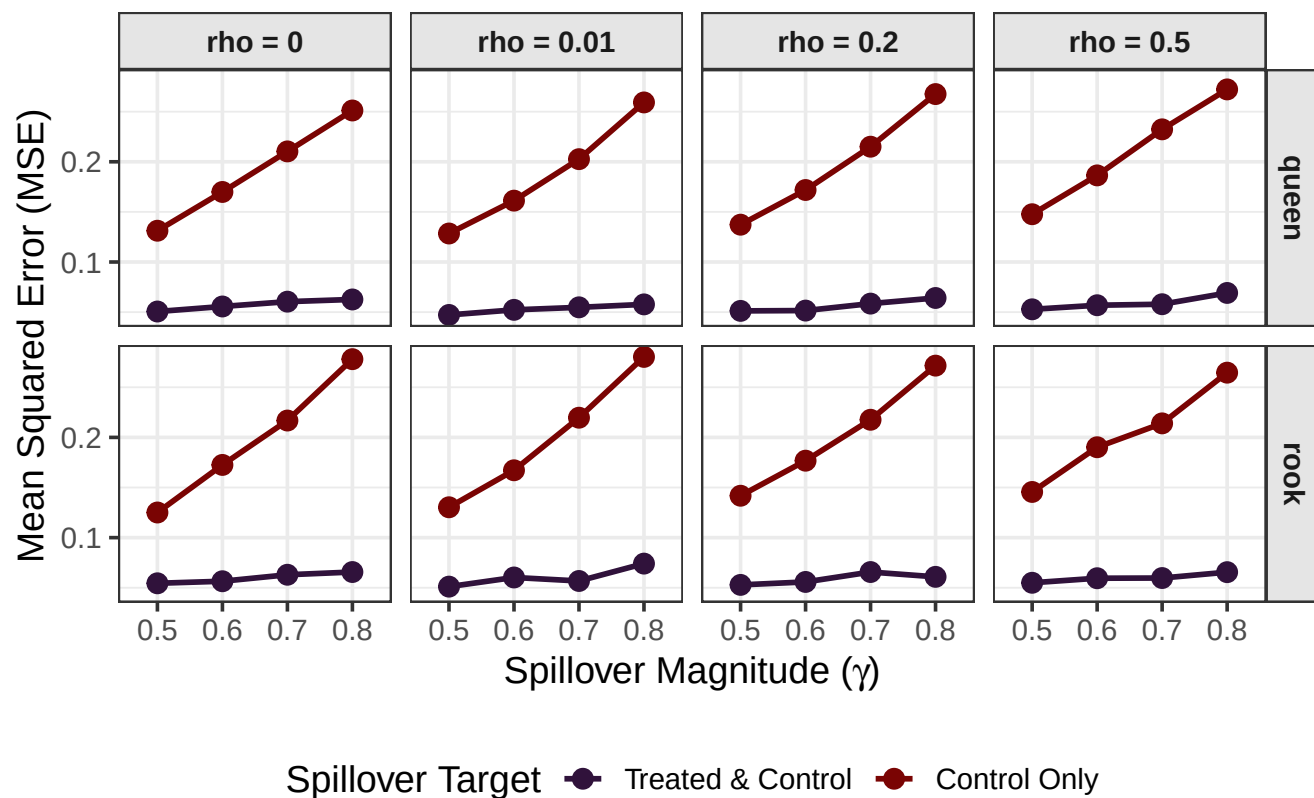


Figure 19: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 6`
```

MSE Dynamics for Design 6

Incidence: iid Uniform

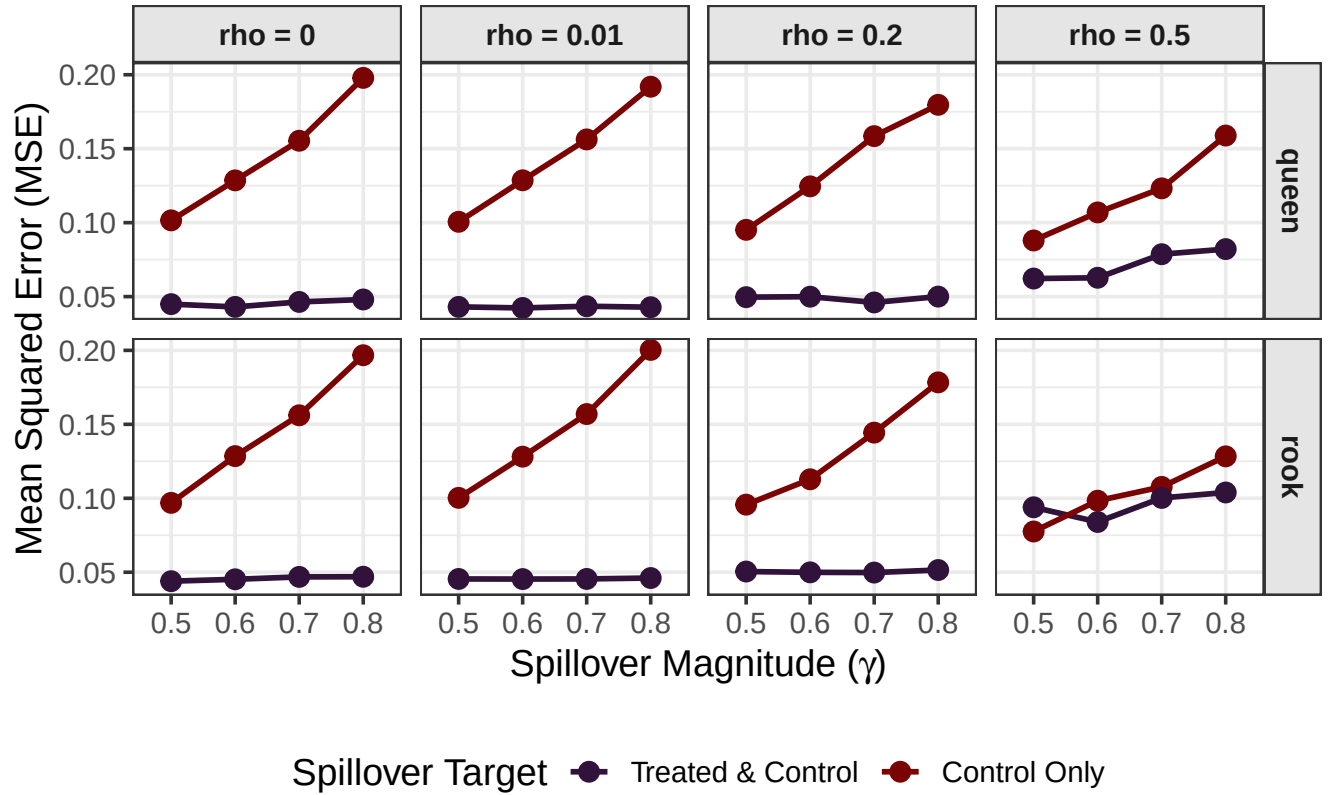


Figure 20: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

7.3 Incidence Mode Comparison

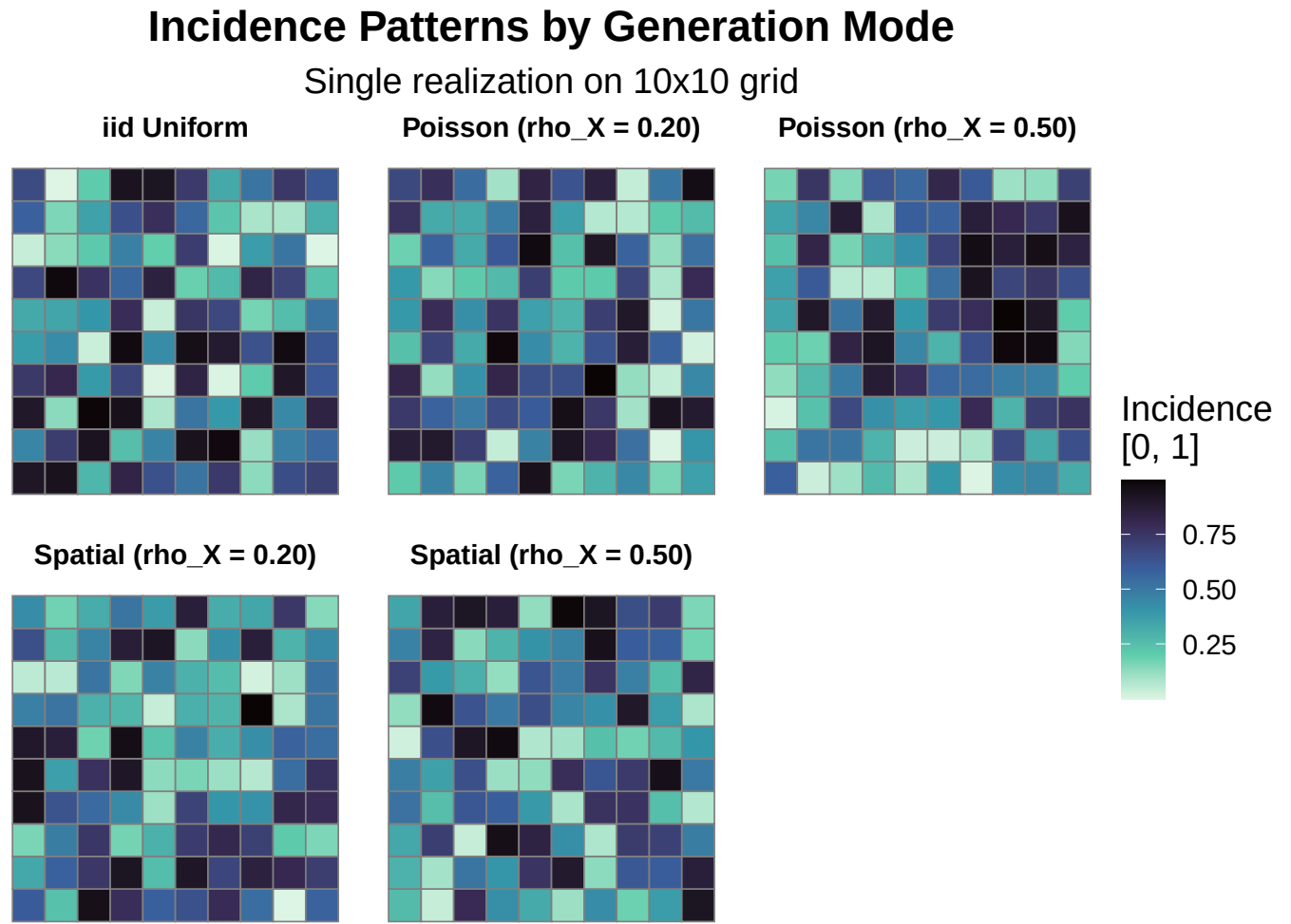


Figure 21: Incidence spatial patterns across the three generation modes for $\rho_X = 0.20$ and 0.50 . Color represents incidence (darker = higher burden). Spatial and Poisson modes show visible spatial clustering absent from iid.

Marginal Distribution of Incidence Values by Mode

Pooled across 50 resamples on 10x10 grid

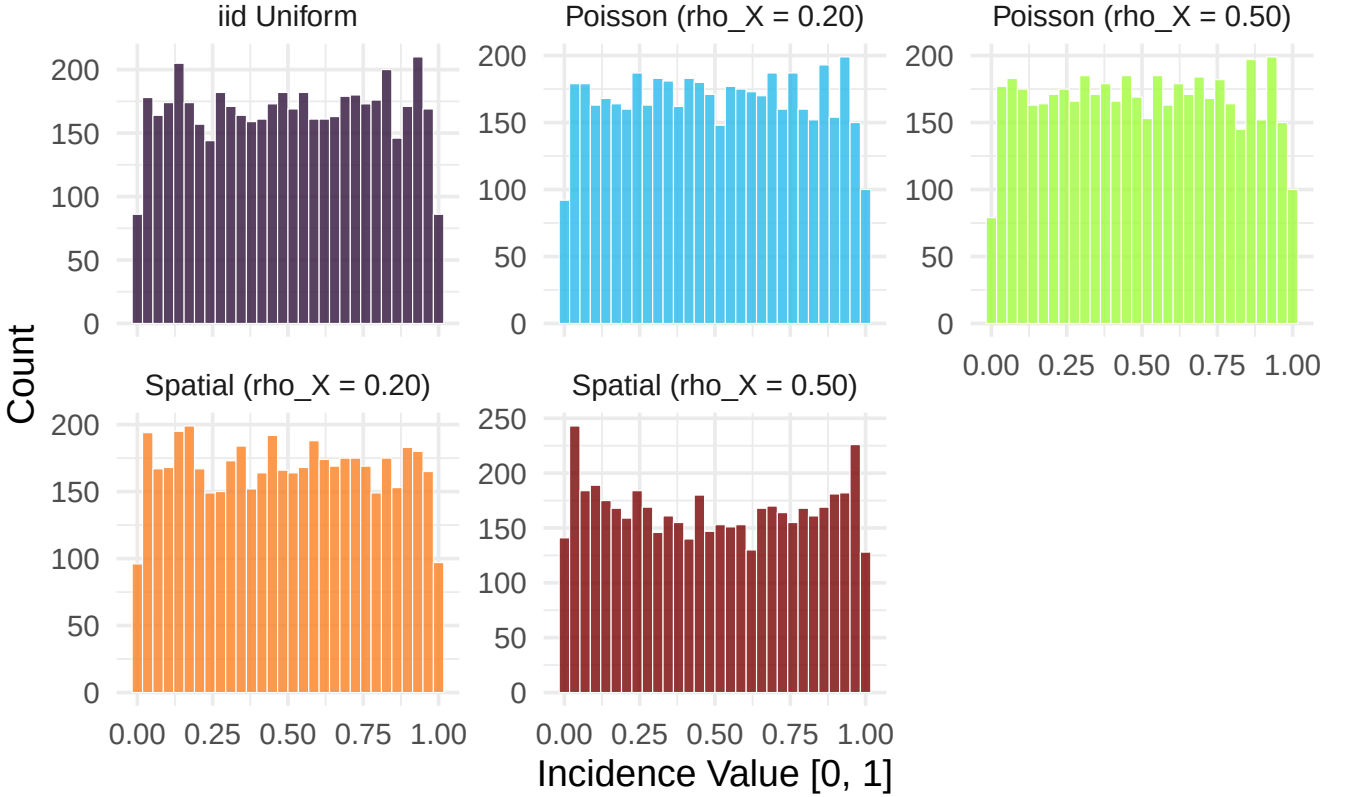


Figure 22: Marginal distribution of incidence values by mode, pooled across 50 resamples on the 10x10 grid. All modes are marginally Uniform on $[0,1]$; spatial structure manifests in spatial covariance, not the marginal.

8 Subgroup Results by Incidence Mode

The following tables report DIM (pre-revision baseline) results separately for each incidence configuration, pooled over rook and queen. Results are **never aggregated across incidence modes**, as each represents a fundamentally different assumption about prior data.

8.1 DIM: iid Uniform

Table 12: DIM (pre-revision) robustness profile: iid Uniform

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0434	0.0574	0.0897	0.1754	0.2812	0.1180
Design 3	0.0458	0.0605	0.0904	0.1306	0.3467	0.0701
Design 6	0.0423	0.0478	0.0945	0.1284	0.2003	0.0806
Design 5	0.0472	0.0575	0.0996	0.1933	0.2803	0.1358

Design 4	0.0775	0.0969	0.1287	0.2295	0.3213	0.1326
Design 1	0.0401	0.0962	0.2441	0.4764	0.7815	0.3801

8.2 DIM: Poisson ($\rho_X = 0.20$)

Table 13: DIM (pre-revision) robustness profile: Poisson ($\rho_X = 0.20$)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0447	0.0569	0.0706	0.0870	0.1132	0.0301
Design 3	0.0467	0.0620	0.0868	0.1307	0.3198	0.0686
Design 6	0.0432	0.0472	0.0905	0.1292	0.1990	0.0819
Design 5	0.0487	0.0568	0.1002	0.1890	0.2745	0.1322
Design 4	0.0792	0.0975	0.1343	0.2251	0.3123	0.1276
Design 1	0.0434	0.0964	0.2541	0.4971	0.7967	0.4007

8.3 DIM: Poisson ($\rho_X = 0.50$)

Table 14: DIM (pre-revision) robustness profile: Poisson ($\rho_X = 0.50$)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0538	0.0716	0.0836	0.1076	0.3393	0.0360
Design 3	0.0480	0.0638	0.0883	0.1318	0.3915	0.0680
Design 6	0.0427	0.0466	0.0902	0.1319	0.2000	0.0854
Design 5	0.0456	0.0571	0.0990	0.1910	0.2739	0.1339
Design 4	0.0795	0.0993	0.1334	0.2189	0.3146	0.1196
Design 1	0.0319	0.0804	0.2761	0.5354	0.7952	0.4550

8.4 DIM: Spatial ($\rho_X = 0.20$)

Table 15: DIM (pre-revision) robustness profile: Spatial ($\rho_X = 0.20$)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0472	0.0675	0.0891	0.1361	0.3716	0.0687
Design 6	0.0436	0.0468	0.0906	0.1270	0.2023	0.0802
Design 2	0.0415	0.0511	0.0941	0.1375	0.2165	0.0865
Design 5	0.0481	0.0593	0.1019	0.1877	0.2834	0.1284
Design 4	0.0776	0.0963	0.1293	0.2338	0.3097	0.1374
Design 1	0.0324	0.0850	0.2532	0.5083	0.7986	0.4234

8.5 DIM: Spatial ($\rho_X = 0.50$)

Table 16: DIM (pre-revision) robustness profile: Spatial ($\rho_X = 0.50$)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0396	0.0667	0.0879	0.1258	0.3491	0.0591
Design 6	0.0417	0.0472	0.0920	0.1280	0.2029	0.0808
Design 3	0.0481	0.0688	0.0927	0.1299	0.3553	0.0610
Design 5	0.0474	0.0561	0.1023	0.1911	0.2841	0.1350
Design 4	0.0774	0.0982	0.1343	0.2289	0.3254	0.1308

9 Tau Sensitivity Analysis

The tau-sweep simulation (12,800 scenarios) swept the direct treatment effect τ across $\{0.8, 1.0, 1.5, 2.0, 3.0\}$. The primary scenario is $\tau = 1.0$; all standard results above use this slice. This section examines whether design rankings and statistical power change across effect sizes. Plots show iid Uniform incidence under queen contiguity (primary). All τ levels reuse the same assignments and noise (common random numbers), so agreement across τ is not independent confirmation.

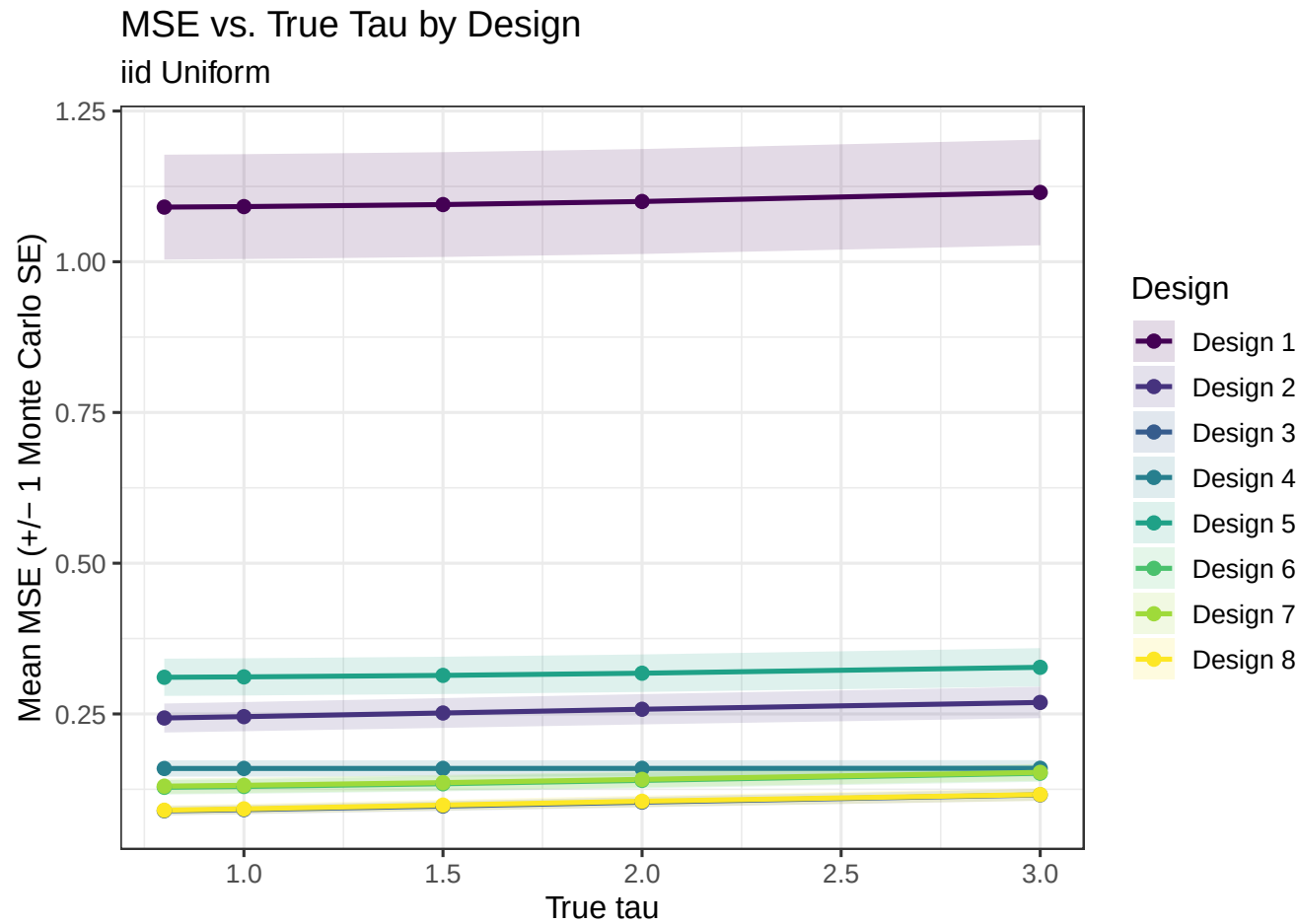


Figure 23: MLE MSE by design across tau values (iid Uniform, queen). Checkerboard has the highest mean MSE at every tau.

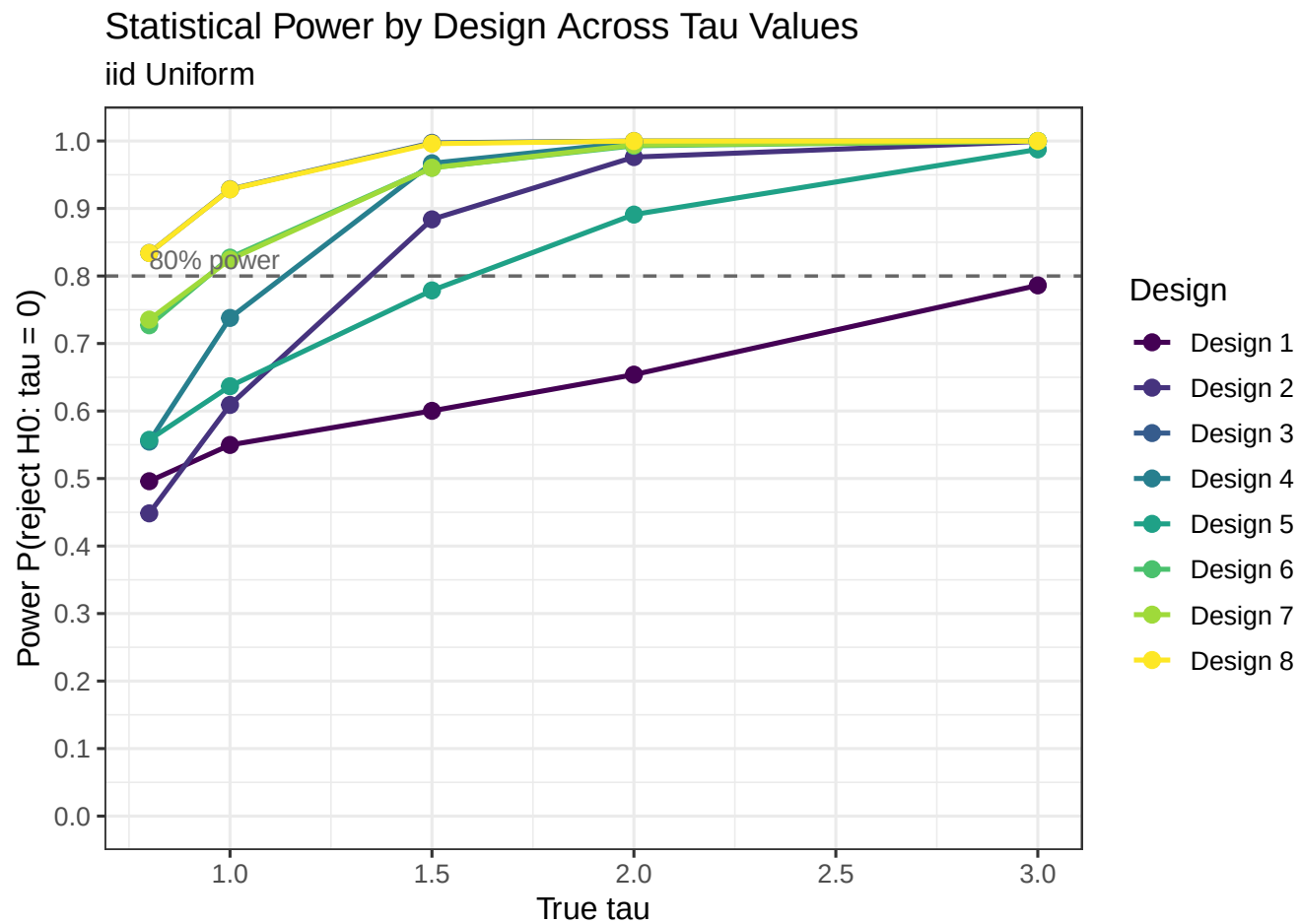


Figure 24: Statistical power $P(\text{reject } H_0: \tau=0)$ by design across τ values (iid Uniform, queen). 80% power reference line shown. Designs with higher MSE require larger τ to achieve adequate power.

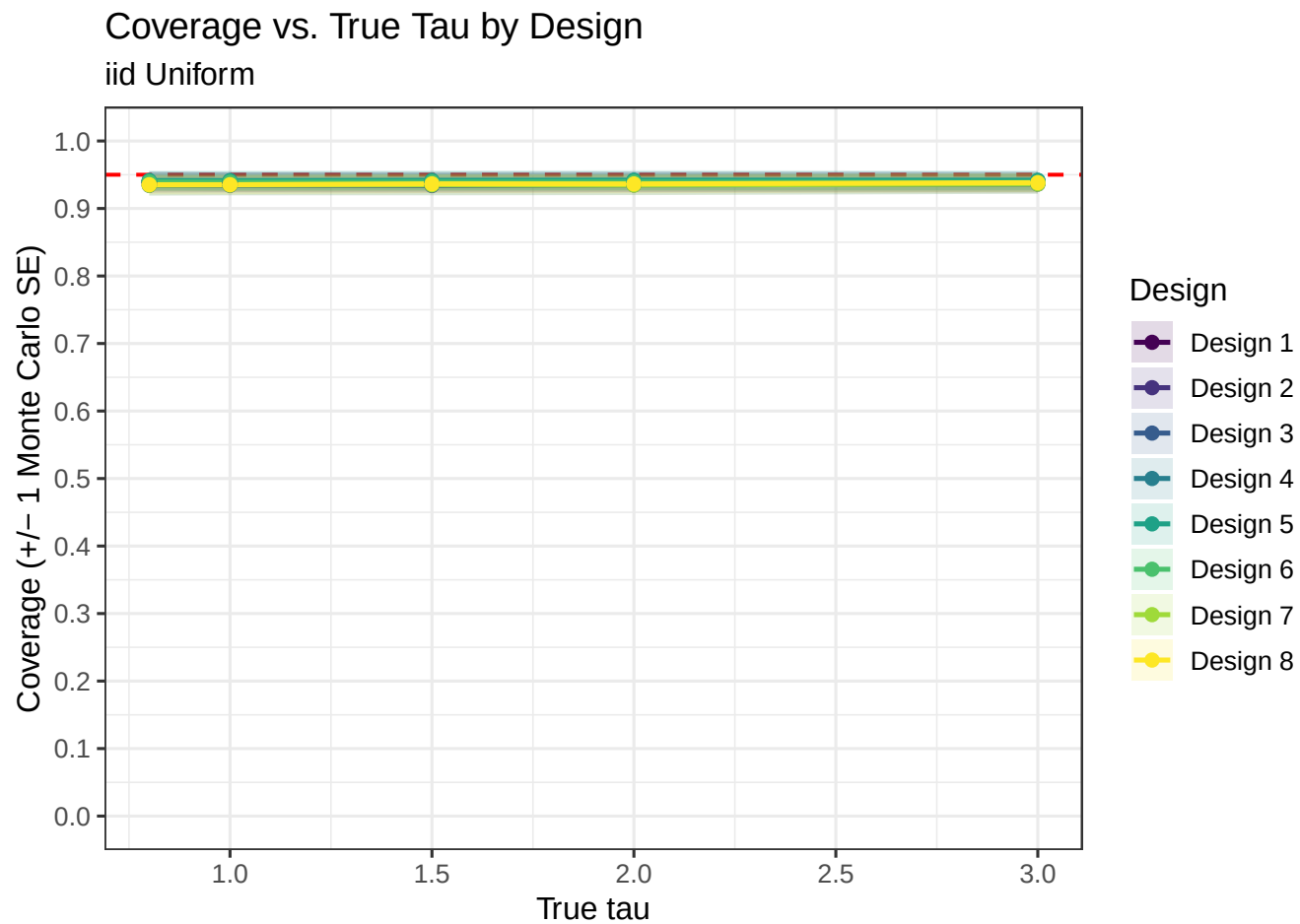


Figure 25: MLE coverage by design across tau values (iid Uniform, queen). Nominal 95% reference line shown. Coverage is near-nominal for every design, including Checkerboard; under rook (not shown) Checkerboard is not identified.

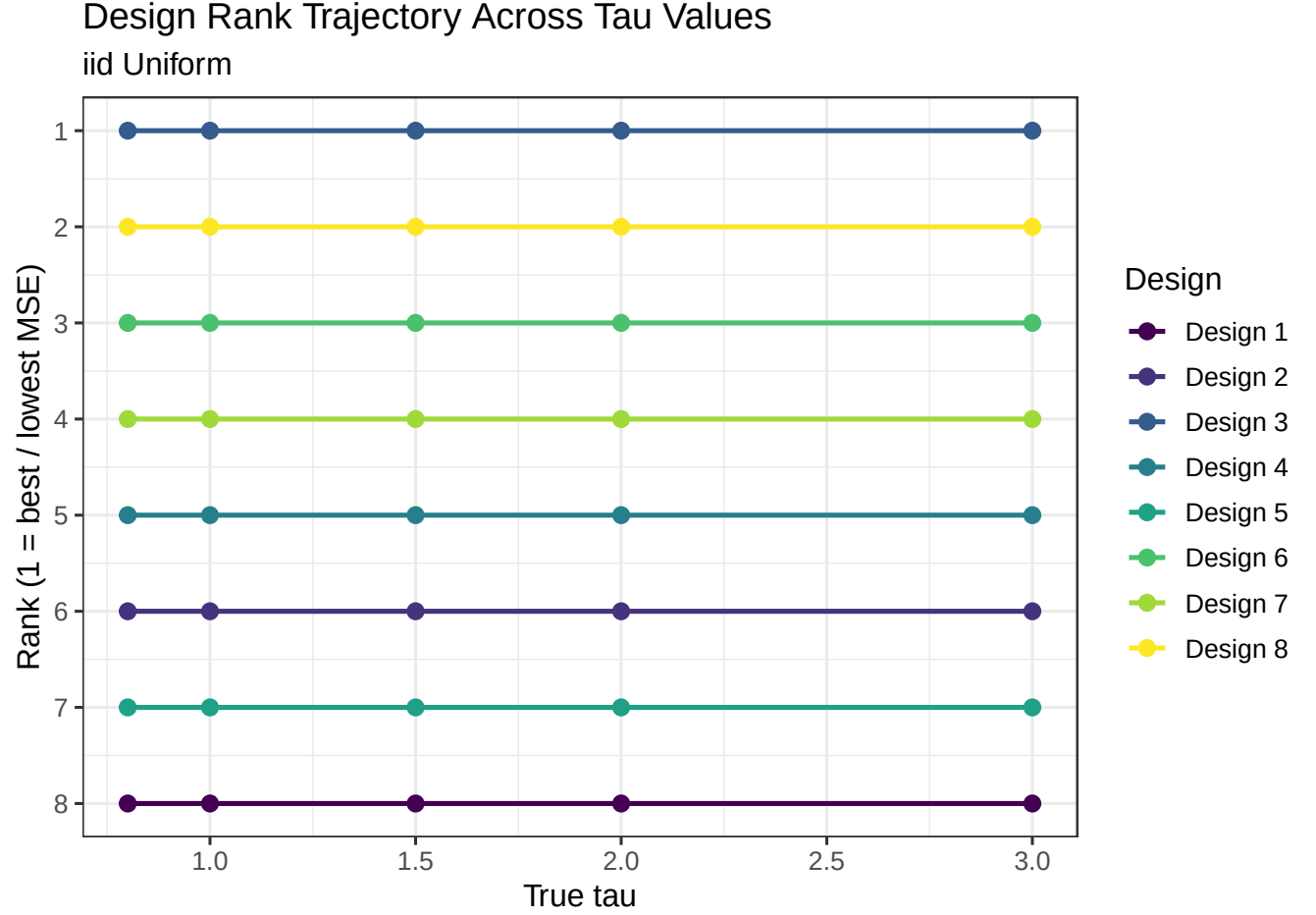


Figure 26: Design rank trajectories across tau values (iid Uniform, queen). Rank 1 = best MSE. Stable trajectories indicate robust design recommendations across effect sizes.

10 Conclusions

10.1 Summary of Findings

The simulation study evaluated 8 treatment assignment designs across 2,560 MLE parameter scenarios (primary $\tau = 1.0$ slice), spanning three incidence generation modes, two contiguity structures, four spatial autocorrelation levels, four spillover magnitudes, and two spillover types. The full tau-sweep covers 12,800 scenarios across $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$.

Key findings (MLE, $\tau = 1.0$; queen primary, rook sensitivity):

1. **Coverage is near-nominal and does not separate the designs.** Mean coverage is 0.936–0.942 for every design under both neighbor types, except Checkerboard under rook. The design differences show up in MSE. The pre-revision DIM baseline under-covers (mean 0.72), but it was run under the old data-generating process and is not a like-for-like comparison.
2. **Saturation Quadrants and Incidence-Guided Saturation Quadrants have the lowest MSE** (queen: 0.090 and 0.091; rook: 0.071 and 0.072). The two are statistically indistinguishable (queen Nemenyi $p = 0.985$, Holm-adjusted Wilcoxon $p = 0.413$; eight-design comparisons in

results/eight_design_supplementary/).

3. **Balanced Halves and Balanced Quartiles form the next tier** (queen: 0.130 and 0.131), and are also statistically indistinguishable (queen Nemenyi $p = 1.000$, Wilcoxon $p = 0.884$). Isolation Buffer (0.160), High Incidence Focus (0.245) and 2x2 Blocking (0.319) follow under queen; under rook, 2x2 Blocking (0.128) and Isolation Buffer (0.133) are close and High Incidence Focus (0.206) is higher.
4. **Checkerboard has the highest MSE.** Under queen its MSE is 1.087, with mean bias 0.001, so almost all of it is variance. Under rook, $WZ = \mathbf{1} - Z$ and τ is not identified (bias -0.652, approximately $-\gamma$; coverage 0.150); these estimates are kept but flagged.
5. **The MSE ordering is the same at every τ under queen** ($\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$), with the caveat that all τ levels share the same assignments and noise.

10.2 Limitations and Planned Extensions

- **Oracle MLE:** The primary MLE model uses the true spillover covariate. A non-oracle fit ($Y \sim Z + X$) was run as a sensitivity analysis (`sim_results_MLEnonoracle_tau_sweep_combined_20260924_025509.rds`) but is not shown here.
- **DIM baseline is pre-revision:** DIM (2,500 fits per scenario) was run under the old data-generating process with 6 designs and was not re-run for the 2026-09 revision.
- **Equal populations:** Poisson mode uses $P_i = 100,000$ for all clusters. Heterogeneous population sizes (log-normal) would more closely match NC county data.
- **Grid sensitivity:** All results use $G = 10$ ($N = 100$). Sensitivity to $G = 8$ or $G = 15$ has not been evaluated.
- **Tau sensitivity:** τ was swept across $\{0.8, 1.0, 1.5, 2.0, 3.0\}$ in the completed tau-sweep run (12,800 scenarios). See Section 9 for power and rank-stability results across effect sizes.

11 References

- Mirzaei, A. et al. (2019). Sudden unexpected death rates. (*SUD base rate 35/100,000*)
- Gan, W. et al. (2019). County-level Poisson regression for SUD mortality.
- Watson, K. et al. (AHA). Census tract-level spatial analysis of SUD.
- LeSage, J. & Pace, R.K. (2009). *Introduction to Spatial Econometrics*. CRC Press.
- Bivand, R., et al. `spdep` R package.
- Bivand, R. & Piras, G. `spatialreg` R package.
- R packages used: `dplyr`, `tidyr`, `ggplot2`, `viridis`, `knitr`, `kableExtra`, `digest`

12 Appendix: Result File Locations

Table 17: Key output files in `results/sim_data/` and `results/` directories.

File	Contents
<code>sim_results_MLE_tau_sweep_combined_20260924_025509.rds</code>	PRIMARY: 12,800 oracle MLE tau-sweep scenarios (2026-09 revision)
<code>sim_results_MLE_tau_sweep_{iid,spatial,poisson}_20260924_025509.rds</code>	Per-incidence-mode splits of the oracle results
<code>sim_results_MLEnonoracle_tau_sweep_combined_20260924_025509.rds</code>	Non-oracle MLE ($Y \sim Z + X$) sensitivity analysis, 12,800 scenarios
<code>surface_results_MLE_tau_sweep_20260924_025509.rds</code>	One row per (scenario, surface); basis of the Monte Carlo SEs
<code>sim_results_DIM_combined_20260304_195321.rds</code>	DIM baseline: 1,920 scenarios (pre-revision DGP, 6-design sweep)
<code>MLE_tau_sweep_design_recommendations_{queen,rook}.pdf</code>	Design recommendation figures + tables (primary tau=1.0)
<code>MLE_tau_sweep_incidence_overview.pdf</code>	Incidence heatmaps and distributions
<code>mle_per_config/MLE_tau_sweep_*.pdf</code> (5 files, per incidence config)	Per-config plot PDFs (primary tau=1.0 slice)
<code>mle_per_config/MLE_tau_sweep_*_tau_sensitivity.pdf</code> (5 files)	Per-config tau sensitivity PDFs (MSE/power/coverage vs tau)

11_statistical_comparisons_report.{html,pdf}	Formal statistical comparisons (Friedman, Nemenyi, Wilcoxon)
00_mathematical_specification.pdf	Mathematical theory document
07_results_summary.pdf	This report
archive/pre_revision_20260924/	Superseded pre-revision (March/April 2026) MLE results and rep

All simulation data files are located in **results/sim_data/**. All rendered reports and PDFs are in **results/** and **paper/report/**.